Mathematical Statistics

Practice problems for the SNU Statistics graduate entrance exam

Part I: 47 original problems (10 points each), solutions from page 11.
Part II: 40 past exam problems (2020–2026, restated), from page 22.

Notation: $\mathrm{Exp}(\theta)$ has mean θ; $\mathrm{Gamma}(\alpha,\beta)$ has shape α and scale β; $\chi^2_\alpha(r)$, $t_\alpha(r)$, $F_\alpha(r_1,r_2)$, z_α are upper-α points. A sequence Y_n is $O_p(1)$ if for every $\epsilon>0$ there are M,N with $P(|Y_n|>M)<\epsilon$ for all $n\ge N$; $Y_n=o_p(1)$ means $Y_n \xrightarrow{p} 0$; $Y_n=O_p(a_n)$ means $Y_n/a_n=O_p(1)$.

Problems

Problem 1 *Analysis* [10 pts]

(a) Using the mean value theorem, show that $\dfrac{x}{1+x} < \log(1+x) < x$ for all $x>0$.

(b) Let $f_n(x)=nxe^{-nx}$ for $x\ge 0$. Find the pointwise limit f. Is the convergence uniform on $[0,\infty)$? On $[a,\infty)$ for a fixed $a>0$?

(c) Show that $\int_0^1 f_n(x)\,dx \to \int_0^1 f(x)\,dx$, even though the convergence in (b) is not uniform on $[0,1]$.

(d) Let $K\subset\mathbb{R}^k$ be compact and $g:K\to\mathbb{R}$ continuous. Prove that g is uniformly continuous on K.

Problem 2 *Joint, marginal and conditional distributions* [10 pts]

(X,Y) is uniformly distributed on $D=\{(x,y): 0<x<1,\ 0<y<x^2\}$.

(a) Find the joint pdf and the marginal pdfs of X and Y.

(b) Find $E(Y\mid X=x)$ and $\mathrm{Var}(Y\mid X=x)$, and use them to compute EY and $\mathrm{Var}\,Y$.

(c) Compute $\mathrm{Cov}(X,Y)$. Are X and Y independent?

(d) Compute $P(Y>X/2)$.

Problem 3 *Transformations* [10 pts]

Let $X,Y \overset{\text{iid}}{\sim} \mathrm{Exp}(1)$.

(a) Find the joint pdf of $U=X/(X+Y)$ and $V=X+Y$. Show that they are independent and identify their distributions.

(b) Using (a), find $E(X\mid X+Y=v)$ and $E\big(X^2\mid X+Y=v\big)$.

(c) Find the pdf of $W=X-Y$.

(d) Let $m=\min(X,Y)$ and $R=\max(X,Y)-\min(X,Y)$. Show that m and R are independent and find their distributions.

Problem 4 *Multivariate normal* [10 pts]

Let $\mathbf{X}=(X_1,X_2,X_3)'\sim N_3(\mathbf{0},\Sigma)$ with $\Sigma=\begin{pmatrix}2&1&0\\1&2&1\\0&1&2\end{pmatrix}$.

(a) Find the distribution of $X_1 - X_2 + X_3$.

(b) Find the constant c for which $X_2 - c(X_1 + X_3)$ and $X_1 + X_3$ are independent, and justify independence.

(c) Find the conditional distribution of X_2 given $(X_1, X_3) = (x_1, x_3)$.

(d) Find the mgf $M(t) = E\exp(tX_1X_3)$ and state where it is finite.

(e) What is the distribution of $\mathbf{X}'\Sigma^{-1}\mathbf{X}$? Justify.

Problem 5 *Stochastic order and limits* **[10 pts]**

(a) Show that if $X_n \xrightarrow{d} X$, then $X_n = O_p(1)$.

(b) Show that if $X_n = O_p(1)$ and $Y_n = o_p(1)$, then $X_nY_n = o_p(1)$.

(c) Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} U(0, \theta)$. Show that $n(\theta - X_{(n)}) \xrightarrow{d} \text{Exp}(\theta)$. Conclude that $X_{(n)} - \theta = O_p(n^{-1})$ and $\sqrt{n}(X_{(n)} - \theta) \xrightarrow{p} 0$.

(d) Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Without quoting the delta method, show $\sqrt{n}(\bar{X}_n^2 - \mu^2) \xrightarrow{d} N(0, 4\mu^2\sigma^2)$, using the decomposition $\bar{X}_n^2 - \mu^2 = 2\mu(\bar{X}_n - \mu) + (\bar{X}_n - \mu)^2$.

Problem 6 *MLE, MME and efficiency* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from the Rayleigh density

$$f(x; \theta) = \frac{x}{\theta}\exp\Big(-\frac{x^2}{2\theta}\Big), \qquad x > 0,\ \theta > 0.$$

(a) Show that $X_1^2 \sim \text{Exp}(2\theta)$. Find the MLE $\hat{\theta}$, and its mean and variance.

(b) Compute the Fisher information $I(\theta)$ and show that $\hat{\theta}$ attains the Cramér–Rao lower bound. Is $\hat{\theta}$ the UMVUE?

(c) Using $EX_1 = \sqrt{\pi\theta/2}$, find the method-of-moments estimator $\tilde{\theta}$ based on $\bar{X}$ and its limiting distribution.

(d) Compute $ARE(\tilde{\theta}, \hat{\theta})$.

Problem 7 *Multinomial reparametrization* **[10 pts]**

A sample of n individuals is classified into three genotypes with counts

$$(N_1, N_2, N_3) \sim \text{Multi}\big(n;\ \theta^2,\ 2\theta(1-\theta),\ (1-\theta)^2\big), \qquad 0 < \theta < 1.$$

(a) Find the MLE $\hat{\theta}$.

(b) Show that $2N_1 + N_2 \sim B(2n, \theta)$, and find $E\hat{\theta}$ and $\text{Var}\,\hat{\theta}$.

(c) Find the Fisher information per individual and the limiting distribution of $\sqrt{n}(\hat{\theta} - \theta)$. Is $\hat{\theta}$ the UMVUE?

(d) With $n = 200$ and $(N_1, N_2, N_3) = (50, 90, 60)$, test at level 0.05 whether the three-cell model above fits, against a general trinomial, using Pearson's statistic. State the degrees of freedom.

Problem 8 *Sufficiency, completeness, UMVUE* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from $f(x;\theta) = e^{-(x-\theta)}$, $x > \theta$, $\theta \in \mathbb{R}$.

(a) Show that $X_{(1)}$ is sufficient, and find its distribution.

(b) Show that $X_{(1)}$ is complete.

(c) Find the UMVUEs of θ and of θ^2.

(d) Show that $X_{(1)}$ and $S = \sum_{i=1}^{n}(X_i - X_{(1)})$ are independent, and find the distribution of S.

Problem 9 *Most powerful and UMP tests* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from the Pareto density $f(x;\theta) = \theta x^{-\theta-1}$, $x > 1$, $\theta > 0$, and let $T = \sum_{i=1}^{n} \log X_i$.

(a) Show that $2\theta T \sim \chi^2(2n)$.

(b) Find the most powerful level-α test of $H_0 : \theta = 1$ vs $H_1 : \theta = \theta_1$ for a fixed $\theta_1 > 1$.

(c) Show that the test in (b) is UMP for $H_0 : \theta \le 1$ vs $H_1 : \theta > 1$, and express its power function with a chi-square cdf.

(d) For $n = 10$ and $\alpha = 0.05$ ($\chi^2_{0.95}(20) = 10.851$), give the rejection region and the power at $\theta = 2$.

Problem 10 *Likelihood ratio test* **[10 pts]**

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Exp}(\theta_1)$ and $Y_1, \dots, Y_m \overset{\text{iid}}{\sim} \text{Exp}(\theta_2)$ be independent samples. Consider $H_0 : \theta_1 = \theta_2$ vs $H_1 : \theta_1 \neq \theta_2$.

(a) Show that $2\log\Lambda$, the LRT statistic, is a function of $B = n\bar{X}/(n\bar{X} + m\bar{Y})$, and describe the form of the rejection region.

(b) Find the exact null distribution of B, and of $\bar{X}/\bar{Y}$.

(c) State the limiting null distribution of $2\log\Lambda$ as $n, m \to \infty$.

Problem 11 *Signed rank test* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from a continuous distribution symmetric about θ, and $W^+ = \sum_i \mathbf{1}\{X_i > 0\} R_i$, where R_i is the rank of $|X_i|$ among $|X_1|, \dots, |X_n|$. Under $H_0 : \theta = 0$:

(a) Show that $W^+ \overset{d}{=} \sum_{j=1}^{n} j B_j$ with $B_j \overset{\text{iid}}{\sim} \text{Bernoulli}(1/2)$. (You may use that the signs and the ranks of $|X_i|$ are independent.)

(b) Find EW^+ and $\operatorname{Var} W^+$, and the exact null distribution of W^+ for $n = 4$.

(c) Show that $W^+ = \sum_{1 \le i \le j \le n} \mathbf{1}\{X_i + X_j > 0\}$.

(d) The ARE of the signed rank test relative to the t test is $12\sigma^2\big(\int f^2\big)^2$, where f is the density of $X_1 - \theta$ and σ^2 its variance. Evaluate it for f uniform on $(-1, 1)$ and for $f(x) = \tfrac{1}{2}e^{-|x|}$.

Problem 12 *Bayes and minimax* **[10 pts]**

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$ with prior $\theta \sim \text{Exp}(\beta)$, i.e. Gamma$(1, \beta)$. Let $T = \sum X_i$ and $s = (n + 1/\beta)^{-1}$.

(a) Find the posterior distribution of θ.

(b) Find the Bayes estimator under squared error loss, and write it as a weighted average of $\bar{X}$ and the prior mean.

(c) Under the loss $L(\theta, a) = (a-\theta)^2/\theta$, show that the Bayes estimator is $a^* = sT$ and that its posterior expected loss is s for every T.

(d) Show that $\bar{X}$ is minimax for the loss in (c).

Problem 13 *Probability inequalities* **[10 pts]**

(a) Using Jensen's inequality, show that $(E|X|^r)^{1/r}$ is nondecreasing in $r > 0$.

(b) Let $\hat{p}$ be the proportion of successes in n Bernoulli(p) trials. Using Chebyshev's inequality, find n that guarantees $P(|\hat{p} - p| \geq 0.05) \leq 0.05$ for every p.

(c) (Chernoff bound) For $X \sim \text{Poisson}(\lambda)$, show $P(X \geq 2\lambda) \leq \inf_{t>0} e^{-2\lambda t} E e^{tX}$, and evaluate the bound.

Problem 14 *Conditional expectation and random sums* **[10 pts]**

Let $N \sim \text{Poisson}(\lambda)$ be independent of $X_1, X_2, \ldots$, which are i.i.d. with mean μ, variance σ^2 and mgf M_X. Let $S = \sum_{i=1}^{N} X_i$ ($S = 0$ if $N = 0$).

(a) Find ES and $\operatorname{Var} S$.

(b) Show that $M_S(t) = \exp\{\lambda(M_X(t) - 1)\}$.

(c) If $X_i \sim \text{Bernoulli}(p)$, identify the distribution of S.

(d) Show that $E(X_1 \mid S, N = n) = S/n$ for $n \geq 1$.

Problem 15 *Moment generating functions and cumulants* **[10 pts]**

(a) Identify the distribution with mgf $(0.3 + 0.7e^t)^5$.

(b) $X_1 \sim \text{Poisson}(2)$ and $X_2 \sim \text{Poisson}(1)$ are independent and $Y = X_1 + 2X_2$. Find the mgf, mean and variance of Y. Is Y Poisson?

(c) Show that the r-th cumulant of Gamma(α, β) is $\alpha\beta^r(r-1)!$, and find its skewness.

(d) A random variable satisfies $EX^k = (k+1)!$ for $k = 0, 1, 2, \ldots$. Identify its distribution.

Problem 16 *Uniform order statistics* **[10 pts]**

Let $U_{(1)} < \cdots < U_{(n)}$ be the order statistics of $n \geq 2$ i.i.d. $U(0,1)$ variables.

(a) Find the joint pdf of $(U_{(1)}, U_{(n)})$.

(b) Show that the range $R = U_{(n)} - U_{(1)}$ has the Beta($n-1, 2$) distribution, and find ER.

(c) Using (b), show that $\operatorname{Cov}(U_{(1)}, U_{(n)}) = \dfrac{1}{(n+1)^2(n+2)}$.

(d) Find $P(U_{(1)} < \frac{1}{2} < U_{(n)})$.

Problem 17 *Normal samples and prediction* **[10 pts]**

Let $X_1, \dots, X_{n+1} \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, and let $\bar{X}$ and S^2 be computed from the first n.

(a) Show that $\dfrac{X_{n+1} - \bar{X}}{S\sqrt{1 + 1/n}} \sim t(n-1)$, and give a 95% prediction interval for X_{n+1}.

(b) Show that $ES = \sigma\sqrt{\dfrac{2}{n-1}}\dfrac{\Gamma(n/2)}{\Gamma((n-1)/2)}$.

(c) Find c such that $c\sum_{i=1}^n (X_i - \bar{X})^2$ minimizes $E\big(c\sum(X_i - \bar{X})^2 - \sigma^2\big)^2$.

Problem 18 *Chi-square, t and F* **[10 pts]**

(a) If $T \sim t(r)$, show $T^2 \sim F(1, r)$.

(b) If $X \sim F(r_1, r_2)$, show $\dfrac{r_1 X}{r_2 + r_1 X} \sim \text{Beta}(r_1/2, r_2/2)$.

(c) $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Exp}(\theta)$ and $Y_1, \dots, Y_m \overset{\text{iid}}{\sim} \text{Exp}(\theta)$ independently. Show $\bar{X}/\bar{Y} \sim F(2n, 2m)$.

(d) Show that $EX = r_2/(r_2 - 2)$ for $X \sim F(r_1, r_2)$, $r_2 > 2$.

Problem 19 *Bivariate normal* **[10 pts]**

(X, Y) is bivariate normal with means 0, variances 1 and correlation $\rho \in (-1, 1)$.

(a) Show that $P(X > 0, Y > 0) = \dfrac{1}{4} + \dfrac{\arcsin \rho}{2\pi}$.

(b) Show that $E\max(X, Y) = \sqrt{(1-\rho)/\pi}$.

(c) Find $E(Y^2 \mid X = x)$ and $\text{Cov}(X^2, Y^2)$.

Problem 20 *Quadratic forms* **[10 pts]**

Let $\mathbf{Z} \sim N_n(\mathbf{0}, I)$.

(a) For $n = 3$, show that $Q = \frac{1}{2}(Z_1 - Z_2)^2 + \frac{1}{6}(Z_1 + Z_2 - 2Z_3)^2 \sim \chi^2(2)$ and that $Q = \sum_{i=1}^3 (Z_i - \bar{Z})^2$.

(b) For symmetric A, show $E(\mathbf{Z}'A\mathbf{Z}) = \text{tr}\, A$ and $\text{Var}(\mathbf{Z}'A\mathbf{Z}) = 2\,\text{tr}(A^2)$.

(c) Let $J = \frac{1}{n}\mathbf{1}\mathbf{1}'$. Show that $\mathbf{Z}'J\mathbf{Z}$ and $\mathbf{Z}'(I - J)\mathbf{Z}$ are independent, and identify their distributions.

Problem 21 *Modes of convergence* **[10 pts]**

(a) Let $P(X_n = n) = 1/n$, $P(X_n = 0) = 1 - 1/n$. Show $X_n \xrightarrow{p} 0$ but $EX_n \not\to 0$.

(b) Let $Z \sim N(0, 1)$ and $X_n = (-1)^n Z$. Show $X_n \xrightarrow{d} Z$ but X_n does not converge in probability to Z.

(c) Show that if $X_n \xrightarrow{d} c$ for a constant c, then $X_n \xrightarrow{p} c$.

(d) Show that if $E(X_n - a)^2 \to 0$, then $X_n \xrightarrow{p} a$.

Problem 22 *Central limit theorem in practice* **[10 pts]**

(a) Approximate $P\big(\sum_{i=1}^{48} U_i > 26\big)$ for $U_i \overset{\text{iid}}{\sim} U(0,1)$.

(b) How large must n be so that a 95% interval $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$ has half-width at most 0.03 regardless of p?

(c) Approximate $P(X \geq 120)$ for $X \sim \text{Poisson}(100)$ with a continuity correction.

Problem 23 *Delta method and variance stabilization* **[10 pts]**

(a) For a Poisson(θ) sample, find the limit law of $\sqrt{n}(\sqrt{\bar{X}} - \sqrt{\theta})$ and use it to build an asymptotic 95% interval for θ.

(b) For an Exp(θ) sample, find the limit law of $\sqrt{n}(\log \bar{X} - \log \theta)$.

(c) For $\hat{p} = X/n$, $X \sim B(n,p)$, find the limit law of $\sqrt{n}(\arcsin\sqrt{\hat{p}} - \arcsin\sqrt{p})$.

Problem 24 *Cauchy location* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from the Cauchy density $\frac{1}{\pi}\{1 + (x-\theta)^2\}^{-1}$.

(a) Using the characteristic function $Ee^{itX} = e^{i\theta t - |t|}$, show that $\bar{X}$ has the same distribution as X_1. Is $\bar{X}$ consistent?

(b) Find the limit law of $\sqrt{n}(\hat{m}_n - \theta)$, where $\hat{m}_n$ is the sample median.

(c) Compute the Fisher information $I(\theta)$ and the ARE of the median relative to an efficient estimator.

Problem 25 *A non-regular scale model* **[10 pts]**

Let $X_1, \dots, X_n$ be a sample from $U[\theta, 2\theta]$, $\theta > 0$.

(a) Find the MLE of θ and a minimal sufficient statistic.

(b) Find an unbiased estimator of θ that is a multiple of the MLE.

(c) Show that $(X_{(1)}, X_{(n)})$ is not complete.

Problem 26 *Two-parameter exponential* **[10 pts]**

Let $X_1, \dots, X_n$ ($n \geq 2$) be a sample from $f(x;\mu,\sigma) = \frac{1}{\sigma} e^{-(x-\mu)/\sigma}$, $x \geq \mu$.

(a) Find the MLEs $\hat{\mu}, \hat{\sigma}$.

(b) Show that $n(\hat{\mu} - \mu)/\sigma \sim \text{Exp}(1)$ and $2n\hat{\sigma}/\sigma \sim \chi^2(2n-2)$, independently.

(c) Given that $(X_{(1)}, \sum X_i)$ is complete sufficient, find the UMVUEs of σ and μ.

Problem 27 *Cramér–Rao bound* **[10 pts]**

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$ and $\eta = e^{-\theta} = P(X_1 = 0)$.

(a) Find the Cramér–Rao lower bound for unbiased estimators of η.

(b) The UMVUE of η is $(1-1/n)^T$, $T=\sum X_i$. Find its variance and show that it exceeds the bound for all $\theta>0$.

(c) Find the limit law of $\sqrt{n}(e^{-\bar X}-e^{-\theta})$ and compare.

Problem 28 *Normal with equal mean and variance* **[10 pts]**

Let $X_1,\dots,X_n$ be a sample from $N(\theta,\theta)$, $\theta>0$.

(a) Show that the MLE is $\hat\theta=\frac{1}{2}\big(-1+\sqrt{1+4m_2}\big)$, where $m_2=\frac{1}{n}\sum X_i^2$.

(b) Compute $I(\theta)$ and the limit law of $\sqrt{n}(\hat\theta-\theta)$.

(c) Compare $\hat\theta$ with $\bar X$ and S^2 by ARE.

Problem 29 *Kullback–Leibler divergence* **[10 pts]**

(a) Compute $KL(\theta_1,\theta_0)=E_{\theta_0}\log\{f(X;\theta_0)/f(X;\theta_1)\}$ for $N(\theta,1)$ and for Poisson(θ).

(b) Using $\log t\le t-1$, show $KL\ge 0$ with equality iff the densities coincide.

(c) Explain why this suggests that the MLE is consistent.

Problem 30 *Rao–Blackwellization* **[10 pts]**

(a) For a Poisson(θ) sample, compute $E\big(\mathbf{1}\{X_1=0\}\mid T\big)$, $T=\sum X_i$, and conclude.

(b) For a Bernoulli(θ) sample ($n\ge 2$), find the UMVUE of θ^2 by conditioning X_1X_2 on T.

(c) For $U(0,\theta)$, improve $2X_1$ by conditioning on $X_{(n)}$.

Problem 31 *Basu's theorem* **[10 pts]**

Let $X_1,\dots,X_n\overset{\text{iid}}{\sim}N(\mu,\sigma^2)$ with σ^2 known.

(a) Use Basu's theorem to show that $\bar X$ is independent of S^2 and of $X_{(n)}-X_{(1)}$.

(b) Show that $E\big(X_{(n)}\mid\bar X\big)=\bar X+c_n\sigma$ for a constant c_n not depending on μ.

(c) Find $E(X_1^2\mid\bar X)$.

Problem 32 *Tests for a normal variance* **[10 pts]**

Let $X_1,\dots,X_n\overset{\text{iid}}{\sim}N(\mu,\sigma^2)$.

(a) With μ known, find the UMP level-α test of $H_0:\sigma^2\le\sigma_0^2$ vs $H_1:\sigma^2>\sigma_0^2$, and its power function.

(b) With μ unknown, show that the LRT of $H_0:\sigma^2=\sigma_0^2$ vs $H_1:\sigma^2\ne\sigma_0^2$ is based on $v=\hat\sigma^2/\sigma_0^2$ through $n(v-1-\log v)$, and describe the rejection region.

Problem 33 *Power and sample size* **[10 pts]**

$X_1,\dots,X_n\overset{\text{iid}}{\sim}N(\mu,4^2)$; test $H_0:\mu\le 10$ vs $H_1:\mu>10$ at $\alpha=0.05$.

(a) Give the UMP test and its power function.

(b) Find the smallest n with power at least 0.9 at $\mu = 12$ ($z_{0.10} = 1.282$).

(c) With that n, compute the power at $\mu = 11$.

(d) Show that no UMP test exists for $H_0 : \mu = 10$ vs $H_1 : \mu \neq 10$.

Problem 34 *LRT for a uniform scale* **[10 pts]**

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} U(0, \theta)$; test $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$.

(a) Find the LRT statistic and the size-α rejection region.

(b) Show that under H_0, $2 \log \Lambda \sim \chi^2(2)$ exactly. Why is this not $\chi^2(1)$?

Problem 35 *Wald, score and LRT* **[10 pts]**

$X \sim B(100, p)$ is observed as $x = 60$; test $H_0 : p = 0.5$ vs $H_1 : p \neq 0.5$ at level 0.05.

(a) Compute the LRT, Wald and score statistics.

(b) State the common limiting null distribution and the decisions.

(c) Which of the three needs only H_0, not the MLE? Give the corresponding 95% confidence set for p (the score interval) as an inequality.

Problem 36 *Pivots and confidence intervals* **[10 pts]**

(a) For an $\text{Exp}(\theta)$ sample, give an exact $1 - \alpha$ confidence interval for θ.

(b) For a $U(0, \theta)$ sample, show that among intervals $[X_{(n)}/b, X_{(n)}/a]$ with $0 < a < b \leq 1$ and coverage $1 - \alpha$, the shortest is $[X_{(n)}, X_{(n)}\alpha^{-1/n}]$.

(c) For $N(\mu, \sigma^2)$ with both unknown, give an exact interval for σ.

Problem 37 *Bayesian prediction and risk* **[10 pts]**

(a) Bernoulli(θ) trials with prior Beta(α, β) give t successes in n. Find $P(X_{n+1} = 1 \mid \mathbf{x})$. What is it for a uniform prior and $t = n$?

(b) $X_1, \dots, X_n \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$, σ^2 known, prior $N(\mu_0, \tau^2)$. Show that the posterior mean has Bayes risk $(n/\sigma^2 + 1/\tau^2)^{-1}$, smaller than the Bayes risk σ^2/n of $\bar{X}$.

(c) In (b), find the frequentist MSE of the posterior mean as a function of θ, and show that it exceeds σ^2/n when $|\theta - \mu_0|$ is large.

Problem 38 *Wilcoxon rank sum* **[10 pts]**

Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be independent samples from continuous distributions, $N = m + n$, and $W = \sum_j R(Y_j)$, the sum of the ranks of the Y's in the pooled sample.

(a) Under H_0 (same distribution), show that every rank R has $ER = \frac{N+1}{2}$, $\text{Var}\, R = \frac{N^2-1}{12}$, and that two distinct ranks have covariance $-\frac{N+1}{12}$.

(b) Deduce $EW = \frac{n(N+1)}{2}$ and $\text{Var}\, W = \frac{mn(N+1)}{12}$.

(c) For $m = n = 3$, find $P(W \geq 14)$ under H_0.

Problem 39 *Analysis: limits and series* **[10 pts]**

(a) Using Problem 1(a), prove $\lim_{n\to\infty}(1+x/n)^n = e^x$ for $x > 0$.

(b) Show that $\sum_{k=0}^{\infty} x^k/k!$ converges uniformly on $[-a, a]$ for every $a > 0$.

(c) Justify differentiating this series term by term, and conclude that its sum $E(x)$ satisfies $E' = E$.

(d) Show that a function with $f'' \geq 0$ on an interval is convex, using the mean value theorem.

Problem 40 *Linear algebra* **[10 pts]**

(a) Show that a symmetric idempotent matrix has eigenvalues 0 or 1 and that its rank equals its trace.

(b) For X of size $n \times p$ with rank p, show that $H = X(X'X)^{-1}X'$ and $I - H$ are symmetric idempotent with traces p and $n - p$, and that $(I - H)X = 0$.

(c) Prove $(I + \mathbf{a}\mathbf{b}')^{-1} = I - \dfrac{\mathbf{a}\mathbf{b}'}{1 + \mathbf{b}'\mathbf{a}}$ when $1 + \mathbf{b}'\mathbf{a} \neq 0$, and $\det(I + \mathbf{a}\mathbf{b}') = 1 + \mathbf{b}'\mathbf{a}$.

(d) Use (c) to invert $\Sigma = (1-\rho)I_k + \rho\mathbf{1}\mathbf{1}'$, and find for which ρ it is positive definite.

Problem 41 *UMVUE for a uniform region* **[10 pts]**

$(X_1, Y_1), \ldots, (X_n, Y_n)$ are i.i.d. uniform on the square $\{(x, y) : |x| + |y| < \theta\}$, $\theta > 0$. Let $W_i = |X_i| + |Y_i|$ and $M = \max_i W_i$.

(a) Find the distribution of W_1, and show that M is sufficient.

(b) Show that M is complete.

(c) Find the UMVUEs of θ and of the area $2\theta^2$.

Problem 42 *Two-parameter uniform* **[10 pts]**

$X_1, \ldots, X_n$ ($n \geq 2$) are i.i.d. $U(\theta_1, \theta_2)$, $\theta_1 < \theta_2$. You may use that $(X_{(1)}, X_{(n)})$ is complete sufficient.

(a) Find $EX_{(1)}$ and $EX_{(n)}$.

(b) Find the UMVUEs of $\theta_2 - \theta_1$, of $(\theta_1 + \theta_2)/2$, and of θ_1.

Problem 43 *UMP test for a uniform location* **[10 pts]**

$X_1, \ldots, X_n$ are i.i.d. $U(\theta, \theta + 1)$. Test $H_0 : \theta = 0$ vs $H_1 : \theta > 0$ at level $\alpha \in (0, 1)$.

(a) For a fixed $\theta_1 \in (0, 1)$, compute the likelihood ratio $p_{\theta_1}(\mathbf{x})/p_0(\mathbf{x})$ on each part of the sample space.

(b) Show that the test rejecting when $X_{(n)} \geq 1$ or $X_{(1)} \geq c$, with $c = 1 - \alpha^{1/n}$, has size α and is UMP.

(c) Find its power function.

Problem 44 *Non-uniqueness of UMP tests* **[10 pts]**

$X_1, \dots, X_n$ are i.i.d. $U(0, \theta)$; test $H_0 : \theta \le \theta_0$ vs $H_1 : \theta > \theta_0$ at level α. Show that both

$$\phi_1 = \mathbf{1}\{X_{(n)} > \theta_0(1-\alpha)^{1/n}\} \qquad \text{and} \qquad \phi_2 = \mathbf{1}\{X_{(n)} > \theta_0\} + \alpha\,\mathbf{1}\{X_{(n)} \le \theta_0\}$$

are UMP, and that they have the same power function.

Problem 45 *LRT on a restricted parameter space* **[10 pts]**

$X_1, \dots, X_n \overset{\text{iid}}{\sim} N(\mu, 1)$ with parameter space $\mu \ge 0$. Test $H_0 : \mu = 0$ vs $H_1 : \mu > 0$.

(a) Find the MLE and show $2\log\Lambda = n(\bar{X}^+)^2$, with $a^+ = \max(a, 0)$.

(b) Show that under H_0, $2\log\Lambda$ has the mixture law $\frac{1}{2}\delta_0 + \frac{1}{2}\chi^2(1)$. Find the size-$\alpha$ critical value ($\alpha < \frac{1}{2}$) and compare with the naive $\chi^2(1)$ calibration.

(c) Find the limit law of $\sqrt{n}(\hat{\mu} - \mu)$ for $\mu > 0$ and for $\mu = 0$.

Problem 46 *Extremes* **[10 pts]**

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Exp}(1)$.

(a) Show that $X_{(n)} - \log n \xrightarrow{d} G$ with $P(G \le x) = \exp(-e^{-x})$ (Gumbel).

(b) Find the exact distribution of $nX_{(1)}$.

(c) Show that $nX_{(1)}$ and $X_{(n)} - \log n$ are asymptotically independent.

Problem 47 *Censored data* **[10 pts]**

Lifetimes $X_i \overset{\text{iid}}{\sim} \text{Exp}(\theta)$ are censored at a fixed time $c > 0$: we observe $Z_i = \min(X_i, c)$ and $\delta_i = \mathbf{1}\{X_i \le c\}$, $i = 1, \dots, n$.

(a) Write the likelihood and show that $(D, T) = (\sum \delta_i, \sum Z_i)$ is sufficient.

(b) Find the MLE of θ (when $D > 0$).

(c) Show that $I_n(\theta) = nP_\theta/\theta^2$ with $P_\theta = 1 - e^{-c/\theta}$, and give the limit law of the MLE. How much information is lost by censoring?

Solutions

Solution 1

(a) Apply the MVT to $\log(1+t)$ on $[0,x]$: $\log(1+x) = \frac{x}{1+\xi}$ for some $\xi \in (0,x)$. Since $1 < 1+\xi < 1+x$, we get $\frac{x}{1+x} < \log(1+x) < x$.

(b) $f_n(0) = 0$, and for $x > 0$, $nxe^{-nx} \to 0$ because $ue^{-u} \to 0$ as $u \to \infty$. So $f \equiv 0$. Since $\frac{d}{dx} f_n = ne^{-nx}(1-nx)$, f_n peaks at $x = 1/n$ with $\sup_{x\ge 0} f_n = f_n(1/n) = e^{-1} \not\to 0$: not uniform on $[0,\infty)$. On $[a,\infty)$, for $n > 1/a$ the function is decreasing there, so $\sup_{x\ge a} f_n = nae^{-na} \to 0$: uniform.

(c) $\int_0^1 nxe^{-nx}dx = \frac{1}{n}\int_0^n ue^{-u}du \le \frac{1}{n}\int_0^\infty ue^{-u}du = \frac{1}{n} \to 0 = \int_0^1 f$.

(d) Suppose not. Then there are $\epsilon > 0$ and $x_n, y_n \in K$ with $|x_n - y_n| < 1/n$ but $|g(x_n) - g(y_n)| \ge \epsilon$. By compactness (Bolzano–Weierstrass), a subsequence $x_{n_k} \to x \in K$, and then $y_{n_k} \to x$ as well. By continuity $g(x_{n_k}) - g(y_{n_k}) \to g(x) - g(x) = 0$, contradicting $\ge \epsilon$.

Solution 2

(a) The area of D is $\int_0^1 x^2 dx = \frac{1}{3}$, so $f(x,y) = 3$ on D. Then $f_X(x) = \int_0^{x^2} 3\,dy = 3x^2$ on $(0,1)$, and $f_Y(y) = \int_{\sqrt{y}}^1 3\,dx = 3(1-\sqrt{y})$ on $(0,1)$.

(b) Given $X = x$, $Y \sim U(0,x^2)$: $E(Y \mid x) = x^2/2$, $\operatorname{Var}(Y \mid x) = x^4/12$. Then $EY = E(X^2)/2 = \frac{1}{2}\cdot\frac{3}{5} = \frac{3}{10}$, and

$$\operatorname{Var} Y = E\Big(\frac{X^4}{12}\Big) + \operatorname{Var}\Big(\frac{X^2}{2}\Big) = \frac{1}{12}\cdot\frac{3}{7} + \frac{1}{4}\Big(\frac{3}{7} - \frac{9}{25}\Big) = \frac{1}{28} + \frac{3}{175} = \frac{37}{700}.$$

(c) $E(XY) = E\{X\,E(Y \mid X)\} = E(X^3)/2 = \frac{1}{2}\cdot\frac{3}{6} = \frac{1}{4}$, $EX = \frac{3}{4}$, so $\operatorname{Cov}(X,Y) = \frac{1}{4} - \frac{3}{4}\cdot\frac{3}{10} = \frac{1}{40}$. Not independent (nonzero covariance; also the support is not a product set).

(d) $\{y > x/2\} \cap D$ is nonempty only where $x^2 > x/2$, i.e. $x > 1/2$:

$$P(Y > X/2) = \int_{1/2}^1 3\Big(x^2 - \frac{x}{2}\Big)dx = 3\Big[\frac{x^3}{3} - \frac{x^2}{4}\Big]_{1/2}^1 = 3\Big(\frac{1}{12} + \frac{1}{48}\Big) = \frac{5}{16}.$$

Solution 3

(a) Inverse map $x = uv$, $y = (1-u)v$, $0 < u < 1$, $v > 0$, with $|J| = |\det \begin{pmatrix} v & u \\ -v & 1-u \end{pmatrix}| = v$. So $f(u,v) = e^{-v}v = \mathbf{1}_{(0,1)}(u)\cdot ve^{-v}$: $U \sim U(0,1)$ and $V \sim \text{Gamma}(2,1)$, independent.

(b) $X = UV$ with U independent of V, so $E(X \mid V = v) = vEU = v/2$ and $E(X^2 \mid V = v) = v^2 EU^2 = v^2/3$.

(c) For $w > 0$, $f_W(w) = \int_0^\infty f_X(y+w)f_Y(y)\,dy = \int_0^\infty e^{-(y+w)}e^{-y}dy = \frac{1}{2}e^{-w}$; by symmetry $f_W(w) = \frac{1}{2}e^{-|w|}$, $w \in \mathbb{R}$ (double exponential).

(d) The order statistics have joint pdf $2e^{-y_1-y_2}$ on $0 < y_1 < y_2$. With $m = y_1$, $r = y_2 - y_1$ ($|J| = 1$), $f(m,r) = 2e^{-2m-r} = 2e^{-2m}\cdot e^{-r}$ on $m, r > 0$. So $m \sim \text{Exp}(1/2)$ and $R \sim \text{Exp}(1)$ are independent (memorylessness).

Solution 4

(a) With $a = (1,-1,1)'$, $\Sigma a = (1,0,1)'$ and $a'\Sigma a = 2$. So $X_1 - X_2 + X_3 \sim N(0,2)$.

(b) $(X_2 - c(X_1+X_3),\ X_1+X_3)$ is a linear image of $\mathbf{X}$, hence bivariate normal, and is independent iff the covariance is 0. $\operatorname{Cov}(X_2, X_1+X_3) = 1+1 = 2$ and $\operatorname{Var}(X_1+X_3) = 2+2+2\cdot 0 = 4$, so $\operatorname{Cov} = 2 - 4c = 0$ gives $c = \frac{1}{2}$.

(c) $\Sigma_{12} = (1,1)$, $\Sigma_{22} = 2I_2$, so $\Sigma_{12}\Sigma_{22}^{-1} = (\frac{1}{2},\frac{1}{2})$. Hence $X_2 \mid (x_1,x_3) \sim N\big(\frac{x_1+x_3}{2},\ 2 - \frac{1}{2}(1+1)\big) = N\big(\frac{x_1+x_3}{2}, 1\big)$.

(d) $\mathrm{Cov}(X_1, X_3) = 0$ and they are jointly normal, so $X_1, X_3 \overset{\text{iid}}{\sim} N(0,2)$. Conditioning on X_3, $E(e^{tX_1X_3} \mid X_3) = \exp(\frac{1}{2} \cdot 2t^2X_3^2) = e^{t^2X_3^2}$. Since $X_3^2/2 \sim \chi^2(1)$, $Ee^{t^2X_3^2} = Ee^{2t^2(X_3^2/2)} = (1-4t^2)^{-1/2}$. So $M(t) = (1-4t^2)^{-1/2}$ for $|t| < \frac{1}{2}$, and $M(t) = \infty$ otherwise.

(e) Σ is positive definite (leading minors $2, 3, 4$). $\mathbf{Z} = \Sigma^{-1/2}\mathbf{X} \sim N_3(\mathbf{0}, I)$ and $\mathbf{X}'\Sigma^{-1}\mathbf{X} = \mathbf{Z}'\mathbf{Z} \sim \chi^2(3)$.

Solution 5

(a) Let F be the cdf of X and $\epsilon > 0$. Since F has at most countably many discontinuities, choose M such that $\pm M$ are continuity points and $P(X > M) + P(X \le -M) < \epsilon/2$. Then

$$P(|X_n| > M) \le 1 - F_n(M) + F_n(-M) \to 1 - F(M) + F(-M) < \epsilon/2,$$

so $P(|X_n| > M) < \epsilon$ for all large n; the finitely many earlier n are handled by enlarging M.

(b) For $\delta, \epsilon > 0$, choose M with $P(|X_n| > M) < \epsilon$ for large n. Then

$$P(|X_nY_n| > \delta) \le P(|X_n| > M) + P(|Y_n| > \delta/M) < \epsilon + o(1).$$

(c) For $0 \le x \le n\theta$, $P(n(\theta - X_{(n)}) > x) = P(X_{(n)} < \theta - x/n) = (1 - \frac{x}{n\theta})^n \to e^{-x/\theta}$. So $n(\theta - X_{(n)}) \xrightarrow{d} \mathrm{Exp}(\theta)$, which is $O_p(1)$ by (a), i.e. $X_{(n)} - \theta = O_p(n^{-1})$. Then $\sqrt{n}(X_{(n)} - \theta) = n^{-1/2} \cdot n(X_{(n)} - \theta) = o(1)O_p(1) = o_p(1)$ by (b).

(d) By the CLT, $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$, so $2\mu\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, 4\mu^2\sigma^2)$. Also $\sqrt{n}(\bar{X}_n - \mu)^2 = \sqrt{n}(\bar{X}_n - \mu) \cdot (\bar{X}_n - \mu) = O_p(1)\, o_p(1) = o_p(1)$, by (a), the WLLN and (b). Slutsky's theorem gives the result.

Solution 6

(a) $P(X^2 > t) = \int_{\sqrt{t}}^{\infty} \frac{x}{\theta} e^{-x^2/2\theta} dx = e^{-t/(2\theta)}$, so $X^2 \sim \mathrm{Exp}(2\theta)$. The log-likelihood is $l(\theta) = \sum \log x_i - n \log\theta - \sum x_i^2/(2\theta)$, with $\dot{l} = -n/\theta + \sum x_i^2/(2\theta^2)$ positive iff $\theta < \sum x_i^2/(2n)$. So $\hat{\theta} = \frac{1}{2n}\sum X_i^2$, with $E\hat{\theta} = \theta$ and $\mathrm{Var}\,\hat{\theta} = \frac{(2\theta)^2}{4n} = \frac{\theta^2}{n}$.

(b) $\partial_\theta^2 \log f = \frac{1}{\theta^2} - \frac{x^2}{\theta^3}$, so $I(\theta) = -\frac{1}{\theta^2} + \frac{2\theta}{\theta^3} = \frac{1}{\theta^2}$. The CR bound for unbiased estimators is $\frac{1}{nI(\theta)} = \frac{\theta^2}{n} = \mathrm{Var}\,\hat{\theta}$. So $\hat{\theta}$ is efficient, hence UMVUE (also: $\sum X_i^2$ is complete sufficient in this exponential family).

(c) $\theta = \frac{2}{\pi}(EX)^2$, so $\tilde{\theta} = \frac{2}{\pi}\bar{X}^2$. $\mathrm{Var}\,X = EX^2 - (EX)^2 = 2\theta - \frac{\pi\theta}{2} = \frac{(4-\pi)\theta}{2}$. With $g(m) = 2m^2/\pi$, $g'(m) = 4m/\pi$ and $m = \sqrt{\pi\theta/2}$, the delta method gives

$$\sqrt{n}(\tilde{\theta} - \theta) \xrightarrow{d} N\Big(0, \frac{16m^2}{\pi^2} \cdot \frac{(4-\pi)\theta}{2}\Big) = N\Big(0, \frac{4(4-\pi)}{\pi}\theta^2\Big).$$

(d) $ARE(\tilde{\theta}, \hat{\theta}) = \dfrac{\theta^2}{4(4-\pi)\theta^2/\pi} = \dfrac{\pi}{4(4-\pi)} \approx 0.915.$

Solution 7

(a) $l(\theta) = (2N_1 + N_2)\log\theta + (N_2 + 2N_3)\log(1-\theta) + \text{const}$, strictly concave, so $\hat{\theta} = \frac{2N_1+N_2}{2n}$.

(b) Each individual contributes $Y_i \in \{2, 1, 0\}$ copies of the allele, with probabilities $\theta^2, 2\theta(1-\theta), (1-\theta)^2$, i.e. $Y_i \sim B(2, \theta)$, independently. So $2N_1 + N_2 = \sum Y_i \sim B(2n, \theta)$, $E\hat{\theta} = \theta$, $\mathrm{Var}\,\hat{\theta} = \frac{\theta(1-\theta)}{2n}$.

(c) Per individual, $I(\theta) = \mathrm{Var}\left(\frac{Y}{\theta} - \frac{2-Y}{1-\theta}\right) = \frac{2\theta(1-\theta)}{\theta^2(1-\theta)^2} = \frac{2}{\theta(1-\theta)}$. Hence $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \theta(1-\theta)/2)$ (here even $\mathrm{Var}\,\hat{\theta} = 1/(nI)$ exactly). $2N_1 + N_2$ is complete sufficient (binomial family), and $\hat{\theta}$ is an unbiased function of it: UMVUE.

(d) $\hat{\theta} = (100 + 90)/400 = 0.475$. Expected counts: $200(0.475)^2 = 45.125$, $200 \cdot 2(0.475)(0.525) = 99.75$, $200(0.525)^2 = 55.125$. Pearson's statistic:

$$\frac{4.875^2}{45.125} + \frac{9.75^2}{99.75} + \frac{4.875^2}{55.125} = 0.527 + 0.953 + 0.431 = 1.91.$$

Degrees of freedom $= \dim\Omega - \dim\Omega_0 = 2-1 = 1$; since $1.91 < \chi^2_{0.05}(1) = 3.84$, the model is not rejected.

Solution 8

(a) $\prod f(x_i;\theta) = e^{-\sum x_i + n\theta}\mathbf{1}\{x_{(1)} > \theta\} = \big[e^{n\theta}\mathbf{1}\{x_{(1)} > \theta\}\big]e^{-\sum x_i}$; by factorization $X_{(1)}$ is sufficient. $P(X_{(1)} > y) = e^{-n(y-\theta)}$ for $y > \theta$, so $X_{(1)} - \theta \sim \mathrm{Exp}(1/n)$.

(b) If $E_\theta g(X_{(1)}) = \int_\theta^\infty g(y) n e^{-n(y-\theta)}dy = 0$ for all θ, then $\int_\theta^\infty g(y)e^{-ny}dy = 0$ for all θ. Differentiating in θ, $g(\theta)e^{-n\theta} = 0$ for almost every θ, so $g = 0$ a.e. and $P_\theta(g(X_{(1)}) = 0) = 1$.

(c) $EX_{(1)} = \theta + \frac{1}{n}$, so $X_{(1)} - \frac{1}{n}$ is the UMVUE of θ (Lehmann–Scheffé). Also $EX^2_{(1)} = \mathrm{Var}\, X_{(1)} + (EX_{(1)})^2 = \frac{1}{n^2} + \theta^2 + \frac{2\theta}{n} + \frac{1}{n^2}$, so

$$E\Big[X^2_{(1)} - \frac{2}{n}X_{(1)}\Big] = \theta^2 + \frac{2\theta}{n} + \frac{2}{n^2} - \frac{2\theta}{n} - \frac{2}{n^2} = \theta^2,$$

and $X^2_{(1)} - \frac{2}{n}X_{(1)}$ is the UMVUE of θ^2.

(d) Write $X_i = \theta + Z_i$ with $Z_i \overset{\text{iid}}{\sim} \mathrm{Exp}(1)$; then $S = \sum(Z_i - Z_{(1)})$ does not depend on θ: it is ancillary. $X_{(1)}$ is complete sufficient, so by Basu's theorem they are independent. With $D_i = (n-i+1)(Z_{(i)} - Z_{(i-1)}) \overset{\text{iid}}{\sim} \mathrm{Exp}(1)$ (exponential spacings), $S = \sum_{i=2}^n (Z_{(i)} - Z_{(1)}) = \sum_{i=2}^n D_i \sim \mathrm{Gamma}(n-1, 1)$.

Solution 9

(a) $P(\log X > t) = P(X > e^t) = e^{-\theta t}$, so $\log X_i \overset{\text{iid}}{\sim} \mathrm{Exp}(1/\theta)$, $T \sim \mathrm{Gamma}(n, 1/\theta)$ and $2\theta T \sim \mathrm{Gamma}(n, 2) = \chi^2(2n)$.

(b) $\dfrac{\prod f(x_i;\theta_1)}{\prod f(x_i;1)} = \theta_1^n \exp\{-(\theta_1 - 1)T\}$, decreasing in T for $\theta_1 > 1$. By Neyman–Pearson, the MP test rejects for $T \le c$. Under $\theta = 1$, $2T \sim \chi^2(2n)$, so $c = \frac{1}{2}\chi^2_{1-\alpha}(2n)$: reject when $2T \le \chi^2_{1-\alpha}(2n)$.

(c) The test does not depend on θ_1, so it is UMP for $\theta = 1$ vs $\theta > 1$. Its power is

$$\gamma(\theta) = P_\theta\big(2T \le \chi^2_{1-\alpha}(2n)\big) = P\big(\chi^2(2n) \le \theta\chi^2_{1-\alpha}(2n)\big),$$

increasing in θ. So $\max_{\theta\le 1}\gamma(\theta) = \gamma(1) = \alpha$, and any level-$\alpha$ test of $\theta \le 1$ is level α for $\theta = 1$; hence the test is UMP for $H_0 : \theta \le 1$. (This is also the one-parameter exponential family result with $g(\theta) = -\theta$ decreasing in θ, applied to T.)

(d) Reject when $2T \le 10.851$, i.e. $\sum \log X_i \le 5.43$. Power at $\theta = 2$: $P(\chi^2(20) \le 21.70) \approx 0.64$.

Solution 10

(a) Unrestricted MLEs: $\hat\theta_1 = \bar X$, $\hat\theta_2 = \bar Y$; under H_0: $\hat\theta_0 = (n\bar X + m\bar Y)/(n+m)$. Since $l(\theta_1,\theta_2) = -n\log\theta_1 - n\bar x/\theta_1 - m\log\theta_2 - m\bar y/\theta_2$, the terms $n\bar x/\hat\theta$ etc. sum to $n+m$ in both cases, and

$$2\log\Lambda = 2\big\{(n+m)\log\hat\theta_0 - n\log\bar X - m\log\bar Y\big\}.$$

With $B = n\bar X/(n\bar X + m\bar Y)$, $\bar X = \frac{(n+m)\hat\theta_0}{n}B$ and $\bar Y = \frac{(n+m)\hat\theta_0}{m}(1-B)$, so

$$2\log\Lambda = -2\{n\log B + m\log(1-B)\} + 2\{n\log\tfrac{n}{n+m} + m\log\tfrac{m}{n+m}\}.$$

The function $-n\log b - m\log(1-b)$ is convex with minimum at $b = n/(n+m)$, so the LRT rejects when $B \le c_1$ or $B \ge c_2$, where $c_1 < \frac{n}{n+m} < c_2$ give equal values of the statistic and total size α.

(b) Under H_0 (common θ), $n\bar X/\theta \sim \mathrm{Gamma}(n,1)$ and $m\bar Y/\theta \sim \mathrm{Gamma}(m,1)$ are independent, so $B \sim \mathrm{Beta}(n,m)$, free of θ. Equivalently $\frac{2n\bar X/\theta}{2n}\Big/\frac{2m\bar Y/\theta}{2m} = \bar X/\bar Y \sim F(2n,2m)$.

(c) $\dim\Omega - \dim\Omega_0 = 2-1 = 1$, so $2\log\Lambda \xrightarrow{d} \chi^2(1)$ under H_0.

Solution 11

(a) Let D_j be the index of the observation whose $|X|$ has rank j. Then $W^+ = \sum_{j=1}^n j\,\mathbf{1}\{X_{D_j} > 0\}$. Since the sign vector is independent of the ranks and its entries are i.i.d. fair, the signs attached to ranks $1, \dots, n$ are again i.i.d. fair, so $W^+ \overset{d}{=} \sum j B_j$.

(b) $EW^+ = \frac{1}{2}\sum j = \frac{n(n+1)}{4}$ and $\operatorname{Var} W^+ = \frac{1}{4}\sum j^2 = \frac{n(n+1)(2n+1)}{24}$. For $n = 4$, W^+ is the sum of a uniformly random subset of $\{1, 2, 3, 4\}$ (16 subsets):

w	0	1	2	3	4	5	6	7	8	9	10
$16\,P(W^+ = w)$	1	1	1	2	2	2	2	2	1	1	1

(c) Order by absolute value: $|Y_{(1)}| < \cdots < |Y_{(n)}|$. Then $W^+ = \sum_i i\mathbf{1}\{Y_{(i)} > 0\} = \sum_i \mathbf{1}\{Y_{(i)} > 0\} + \sum_{j<i} \mathbf{1}\{Y_{(i)} > 0\}$, and for $j < i$, $Y_{(i)} > 0 \iff Y_{(i)} + Y_{(j)} > 0$ since $|Y_{(j)}| < |Y_{(i)}|$. Hence $W^+ = \sum_{i \le j} \mathbf{1}\{X_i + X_j > 0\}$.

(d) Uniform on $(-1, 1)$: $\int f^2 = \frac{1}{2}$, $\sigma^2 = \frac{1}{3}$, so $ARE = 12 \cdot \frac{1}{3} \cdot \frac{1}{4} = 1$. Double exponential: $\int f^2 = \int \frac{1}{4}e^{-2|x|} = \frac{1}{4}$, $\sigma^2 = 2$, so $ARE = 12 \cdot 2 \cdot \frac{1}{16} = \frac{3}{2}$.

Solution 12

(a) $\pi(\theta \mid \mathbf{x}) \propto e^{-n\theta}\theta^T \cdot e^{-\theta/\beta} = \theta^T e^{-\theta/s}$, so $\theta \mid \mathbf{x} \sim \text{Gamma}(T + 1, s)$.

(b) The posterior mean is $s(T + 1) = \dfrac{n}{n + 1/\beta}\bar{X} + \dfrac{1/\beta}{n + 1/\beta}\beta$, a weighted average of $\bar{X}$ and the prior mean β.

(c) The posterior expected loss is $\rho(a) = a^2 E(\theta^{-1} \mid \mathbf{x}) - 2a + E(\theta \mid \mathbf{x})$. For $T \ge 1$, $E(\theta^{-1} \mid \mathbf{x}) = \frac{1}{sT}$, so ρ is minimized at $a^* = sT$ with $\rho(a^*) = s(T + 1) - sT = s$. For $T = 0$, $E(\theta^{-1} \mid \mathbf{x}) = \infty$, so only $a = 0 = sT$ has finite loss, $\rho(0) = E(\theta \mid \mathbf{x}) = s$. In all cases the Bayes estimator is sT and its posterior expected loss is s, so its Bayes risk is $r(\pi_\beta) = s = (n + 1/\beta)^{-1}$.

(d) $\bar{X}$ has risk $E\big[(\bar{X} - \theta)^2/\theta\big] = \frac{\theta/n}{\theta} = \frac{1}{n}$ for all θ. For any estimator δ, $\max_\theta R_\delta(\theta) \ge r(\pi_\beta, \delta) \ge (n + 1/\beta)^{-1}$ for every β; letting $\beta \to \infty$, $\max_\theta R_\delta \ge \frac{1}{n} = \max_\theta R_{\bar{X}}$. So $\bar{X}$ is minimax (limit of Bayes estimators, least favourable sequence of priors).

Solution 13

(a) For $0 < r < s$, $u \mapsto u^{s/r}$ is convex on $[0, \infty)$, so by Jensen $E|X|^s = E\{(|X|^r)^{s/r}\} \ge (E|X|^r)^{s/r}$. Take the $1/s$ power.

(b) $\operatorname{Var}\hat{p} = p(1 - p)/n \le 1/(4n)$, so $P(|\hat{p} - p| \ge 0.05) \le \frac{1}{4n(0.05)^2} = \frac{100}{n} \le 0.05$ once $n \ge 2000$.

(c) For $t > 0$, Markov's inequality applied to e^{tX} gives $P(X \ge 2\lambda) \le e^{-2\lambda t} E e^{tX} = \exp\{-2\lambda t + \lambda(e^t - 1)\}$. The exponent is minimized at $e^t = 2$, giving $P(X \ge 2\lambda) \le e^{\lambda(1 - 2\log 2)} = (e/4)^\lambda$.

Solution 14

(a) $E(S \mid N) = N\mu$ and $\operatorname{Var}(S \mid N) = N\sigma^2$, so $ES = \lambda\mu$ and $\operatorname{Var} S = E(N\sigma^2) + \operatorname{Var}(N\mu) = \lambda(\sigma^2 + \mu^2)$.

(b) $E(e^{tS} \mid N) = M_X(t)^N$, and $Es^N = e^{\lambda(s-1)}$ for Poisson N; so $M_S(t) = \exp\{\lambda(M_X(t) - 1)\}$.

(c) $M_X(t) = 1 - p + pe^t$ gives $M_S(t) = \exp\{\lambda p(e^t - 1)\}$: $S \sim \text{Poisson}(\lambda p)$ (thinning).

(d) Given $N = n$, $(X_1, \dots, X_n)$ is exchangeable, so $E(X_i \mid S, N = n)$ is the same for all $i \le n$; these sum to $E(S \mid S, N = n) = S$. Hence each equals S/n.

Solution 15

(a) $B(5, 0.7)$.

(b) $M_Y(t) = M_{X_1}(t)M_{X_2}(2t) = \exp\{2(e^t - 1) + (e^{2t} - 1)\}$, $EY = 2 + 2 \cdot 1 = 4$, $\operatorname{Var} Y = 2 + 4 \cdot 1 = 6$. Not Poisson, since its mean and variance differ.

(c) $K(t) = -\alpha\log(1-\beta t) = \alpha\sum_{r\ge1}\frac{(\beta t)^r}{r}$, so $\kappa_r = r!\cdot\frac{\alpha\beta^r}{r} = \alpha\beta^r(r-1)!$. Skewness $= \kappa_3/\kappa_2^{3/2} = 2\alpha\beta^3/(\alpha\beta^2)^{3/2} = 2/\sqrt{\alpha}$.

(d) $M(t) = \sum_k EX^k\frac{t^k}{k!} = \sum_k (k+1)t^k = (1-t)^{-2}$ for $|t|<1$. This is the Gamma$(2,1)$ mgf, and an mgf finite near 0 determines the distribution.

Solution 16

(a) $f(u,v) = n(n-1)(v-u)^{n-2}$ on $0<u<v<1$.

(b) With $r = v-u$ and $u\in(0,1-r)$, $f_R(r) = \int_0^{1-r} n(n-1)r^{n-2}du = n(n-1)r^{n-2}(1-r)$, the Beta$(n-1,2)$ density $(1/B(n-1,2) = n(n-1))$. $ER = \frac{n-1}{n+1}$.

(c) $U_{(1)}\sim$ Beta$(1,n)$ and $U_{(n)}\sim$ Beta$(n,1)$ both have variance $\frac{n}{(n+1)^2(n+2)}$, and $\operatorname{Var} R = \frac{2(n-1)}{(n+1)^2(n+2)}$. From $\operatorname{Var} R = \operatorname{Var} U_{(n)} + \operatorname{Var} U_{(1)} - 2\operatorname{Cov}$,

$$\operatorname{Cov}(U_{(1)},U_{(n)}) = \frac{2n-2(n-1)}{2(n+1)^2(n+2)} = \frac{1}{(n+1)^2(n+2)}.$$

(d) $1 - P(\text{all} < \frac12) - P(\text{all} > \frac12) = 1 - 2^{1-n}$.

Solution 17

(a) $X_{n+1} - \bar X \sim N(0,\sigma^2(1+1/n))$, and it is independent of S^2 (X_{n+1} is independent of the first n, and $\bar X$ is independent of S^2). Dividing by $S\sqrt{1+1/n}$ gives $Z/\sqrt{V/(n-1)}$ with $V = (n-1)S^2/\sigma^2\sim\chi^2(n-1)$: a $t(n-1)$ variable. Prediction interval: $\bar X \pm t_{0.025}(n-1)S\sqrt{1+1/n}$.

(b) With $V\sim\chi^2(n-1)$,

$$E\sqrt{V} = \int_0^\infty \frac{v^{n/2-1}e^{-v/2}}{\Gamma(\frac{n-1}{2})2^{(n-1)/2}}dv = \frac{\Gamma(\frac n2)2^{n/2}}{\Gamma(\frac{n-1}{2})2^{(n-1)/2}} = \sqrt2\frac{\Gamma(n/2)}{\Gamma((n-1)/2)},$$

and $ES = \frac{\sigma}{\sqrt{n-1}}E\sqrt{V}$.

(c) With $\sum(X_i-\bar X)^2 = \sigma^2V$, $EV = n-1$, $EV^2 = 2(n-1)+(n-1)^2$,

$$E(c\sigma^2V-\sigma^2)^2 = \sigma^4\{c^2(n-1)(n+1) - 2c(n-1)+1\},$$

minimized at $c = \frac{1}{n+1}$.

Solution 18

(a) $T^2 = Z^2/(V/r)$ with $Z^2\sim\chi^2(1)$ independent of $V\sim\chi^2(r)$: this is $F(1,r)$.

(b) Write $X = \frac{V_1/r_1}{V_2/r_2}$. Then $\frac{r_1X}{r_2+r_1X} = \frac{V_1}{V_1+V_2}$ with independent $V_i\sim$ Gamma$(r_i/2,2)$, which is Beta$(r_1/2,r_2/2)$.

(c) $2n\bar X/\theta\sim\chi^2(2n)$ and $2m\bar Y/\theta\sim\chi^2(2m)$ are independent, and $\bar X/\bar Y = \frac{(2n\bar X/\theta)/(2n)}{(2m\bar Y/\theta)/(2m)}\sim F(2n,2m)$.

(d) $EX = \frac{r_2}{r_1}EV_1\,EV_2^{-1} = \frac{r_2}{r_1}\cdot r_1\cdot\frac{1}{r_2-2}$, since $EV^{-1} = \frac{\Gamma(r/2-1)2^{r/2-1}}{\Gamma(r/2)2^{r/2}} = \frac{1}{r-2}$ for $V\sim\chi^2(r)$.

Solution 19

(a) Write $Y = \rho X + \sqrt{1-\rho^2}Z$ with X,Z i.i.d. $N(0,1)$, and $X = R\cos\Theta$, $Z = R\sin\Theta$ with $\Theta\sim U(-\pi,\pi)$ independent of R. With $\phi = \arcsin\rho$, $Y = R\sin(\Theta+\phi)$. So $\{X>0, Y>0\} = \{\Theta\in(-\frac\pi2,\frac\pi2)\cap(-\phi,\pi-\phi)\} = \{\Theta\in(-\phi,\frac\pi2)\}$, of probability $\frac{\pi/2+\phi}{2\pi} = \frac14 + \frac{\arcsin\rho}{2\pi}$.

(b) $\max(X,Y) = \frac{X+Y}{2} + \frac{|X-Y|}{2}$, and $X-Y\sim N(0,2(1-\rho))$ with $E|X-Y| = \sqrt{2(1-\rho)}\sqrt{2/\pi}$. Hence $E\max(X,Y) = \sqrt{(1-\rho)/\pi}$.

(c) $Y\mid X=x\sim N(\rho x, 1-\rho^2)$, so $E(Y^2\mid x) = \rho^2x^2+1-\rho^2$. Then $EX^2Y^2 = E\{X^2(\rho^2X^2+1-\rho^2)\} = 3\rho^2+1-\rho^2$ and $\operatorname{Cov}(X^2,Y^2) = 2\rho^2$.

Solution 20

(a) $\mathbf{a} = (1,-1,0)'/\sqrt{2}$ and $\mathbf{b} = (1,1,-2)'/\sqrt{6}$ are orthonormal, so $U = \mathbf{a}'\mathbf{Z}$, $V = \mathbf{b}'\mathbf{Z}$ are i.i.d. $N(0,1)$ and $Q = U^2 + V^2 \sim \chi^2(2)$. Together with $\mathbf{1}/\sqrt{3}$ they form an orthonormal basis, so $\sum Z_i^2 = 3\bar{Z}^2 + U^2 + V^2$, i.e. $Q = \sum Z_i^2 - 3\bar{Z}^2 = \sum (Z_i - \bar{Z})^2$.

(b) Write $A = P\Lambda P'$ with P orthogonal and $\mathbf{Y} = P'\mathbf{Z} \sim N(\mathbf{0}, I)$. Then $\mathbf{Z}'A\mathbf{Z} = \sum \lambda_i Y_i^2$, with mean $\sum \lambda_i = \operatorname{tr} A$ and variance $\sum \lambda_i^2 \operatorname{Var}(Y_i^2) = 2\sum \lambda_i^2 = 2 \operatorname{tr} A^2$.

(c) J and $I - J$ are symmetric idempotent and $J(I-J) = 0$, so $J\mathbf{Z}$ and $(I-J)\mathbf{Z}$ are uncorrelated jointly normal vectors, hence independent; the quadratic forms are their squared lengths. Since $\operatorname{tr} J = 1$ and $\operatorname{tr}(I-J) = n-1$, they are $\chi^2(1)$ and $\chi^2(n-1)$.

Solution 21

(a) $P(|X_n| > \epsilon) = 1/n \to 0$, but $EX_n = n \cdot \frac{1}{n} = 1$ for all n.

(b) By symmetry, $X_n \sim N(0,1)$ for every n, so $X_n \xrightarrow{d} Z$ trivially. For odd n, $X_n - Z = -2Z$ and $P(|2Z| > \epsilon)$ is a fixed positive number, so $X_n \not\xrightarrow{p} Z$.

(c) For $\epsilon > 0$, $c - \epsilon$ and $c + \epsilon/2$ are continuity points of the limit cdf $\mathbf{1}\{x \geq c\}$, so

$$P(|X_n - c| \geq \epsilon) \leq P(X_n \leq c - \epsilon) + 1 - P(X_n \leq c + \epsilon/2) \to 0 + 0.$$

(d) Markov's inequality: $P(|X_n - a| \geq \epsilon) \leq E(X_n - a)^2/\epsilon^2 \to 0$.

Solution 22

(a) The sum has mean 24 and variance $48/12 = 4$, so $P \approx 1 - \Phi\big(\frac{26-24}{2}\big) = 1 - \Phi(1) = 0.159$.

(b) Since $p(1-p) \leq \frac{1}{4}$, we need $1.96\sqrt{0.25/n} \leq 0.03$, i.e. $n \geq (1.96/0.06)^2 = 1067.1$, so $n = 1068$.

(c) $P(X \geq 120) = P(X > 119.5) \approx 1 - \Phi\big(\frac{19.5}{10}\big) = 1 - \Phi(1.95) = 0.026$.

Solution 23

(a) $g(\theta) = \sqrt{\theta}$, $g'(\theta) = \frac{1}{2\sqrt{\theta}}$, and $\sqrt{n}(\bar{X} - \theta) \xrightarrow{d} N(0,\theta)$, so $\sqrt{n}(\sqrt{\bar{X}} - \sqrt{\theta}) \xrightarrow{d} N(0, \frac{1}{4})$, free of θ. Hence $\sqrt{\theta} \in \sqrt{\bar{X}} \pm \frac{1.96}{2\sqrt{n}}$, and squaring (lower end truncated at 0) gives the interval for θ.

(b) $g = \log$, $g'(\theta) = 1/\theta$, variance $\theta^2 \cdot \theta^{-2} = 1$: the limit is $N(0,1)$.

(c) $g(p) = \arcsin\sqrt{p}$ has $g'(p) = \frac{1}{2\sqrt{p(1-p)}}$, so the limit is $N\big(0, \frac{p(1-p)}{4p(1-p)}\big) = N(0, \frac{1}{4})$.

Solution 24

(a) $Ee^{it\bar{X}} = \{\varphi(t/n)\}^n = e^{i\theta t - |t|}$, so $\bar{X} \overset{d}{=} X_1$. Then $P(|\bar{X} - \theta| > \epsilon) = P(|X_1 - \theta| > \epsilon)$ does not tend to 0: $\bar{X}$ is not consistent.

(b) The median of X_1 is θ with $f(\theta) = 1/\pi$, so $\sqrt{n}(\hat{m}_n - \theta) \xrightarrow{d} N(0, \frac{1}{4f(\theta)^2}) = N(0, \frac{\pi^2}{4})$.

(c) The score is $\frac{2(x-\theta)}{1+(x-\theta)^2}$, so with Y standard Cauchy,

$$I(\theta) = E\frac{4Y^2}{(1+Y^2)^2} = \frac{4}{\pi}\int_{-\infty}^{\infty} \frac{y^2}{(1+y^2)^3} dy = \frac{4}{\pi} \cdot \frac{\pi}{8} = \frac{1}{2},$$

using $y = \tan u$: $\int \sin^2 u \cos^2 u \, du = \frac{\pi}{8}$ over $(-\frac{\pi}{2}, \frac{\pi}{2})$. The efficient limiting variance is $1/I = 2$, so $ARE(\text{median}, \text{MLE}) = \frac{2}{\pi^2/4} = \frac{8}{\pi^2} \approx 0.81$.

Solution 25

(a) $L(\theta) = \theta^{-n}\mathbf{1}\{x_{(n)}/2 \leq \theta \leq x_{(1)}\}$, decreasing on its support, so $\hat{\theta} = X_{(n)}/2$. L depends on the data only through $(x_{(1)}, x_{(n)})$, and two samples give proportional likelihoods only if these agree, so $(X_{(1)}, X_{(n)})$ is minimal sufficient.

(b) $X_i = \theta(1+U_i)$ with $U_i \overset{\text{iid}}{\sim} U(0,1)$, so $EX_{(n)} = \theta\frac{2n+1}{n+1}$ and $EX_{(1)} = \theta\frac{n+2}{n+1}$. Thus $\frac{n+1}{2n+1}X_{(n)} = \frac{2(n+1)}{2n+1}\hat\theta$ is unbiased.

(c) $g = (2n+1)X_{(1)} - (n+2)X_{(n)}$ has $E_\theta g = \frac{\theta}{n+1}\{(2n+1)(n+2) - (n+2)(2n+1)\} = 0$ for all θ, but $P(g=0) = 0$. So the statistic is not complete.

Solution 26

(a) $L = \sigma^{-n}\exp\{-\sum(x_i-\mu)/\sigma\}\mathbf{1}\{\mu \le x_{(1)}\}$ increases in μ up to $x_{(1)}$, so $\hat\mu = X_{(1)}$. Then $\log L = -n\log\sigma - \sum(x_i - x_{(1)})/\sigma$ is maximized at $\hat\sigma = \bar X - X_{(1)}$.

(b) $X_i = \mu + \sigma Z_i$ with $Z_i \overset{\text{iid}}{\sim} \text{Exp}(1)$. The normalized spacings $D_i = (n-i+1)(Z_{(i)} - Z_{(i-1)})$ are i.i.d. Exp(1). So $n(\hat\mu-\mu)/\sigma = D_1 \sim \text{Exp}(1)$, and $n\hat\sigma/\sigma = \sum(Z_i - Z_{(1)}) = \sum_{i\ge 2} D_i \sim \text{Gamma}(n-1,1)$, i.e. $2n\hat\sigma/\sigma \sim \chi^2(2n-2)$, independent of D_1.

(c) $E(n\hat\sigma) = (n-1)\sigma$, so $\tilde\sigma = \frac{n\hat\sigma}{n-1} = \frac{1}{n-1}\sum(X_i - X_{(1)})$ is the UMVUE of σ. Since $EX_{(1)} = \mu + \sigma/n$, the UMVUE of μ is $X_{(1)} - \tilde\sigma/n$.

Solution 27

(a) $I(\theta) = 1/\theta$, so the bound is $\frac{(de^{-\theta}/d\theta)^2}{nI(\theta)} = \frac{\theta e^{-2\theta}}{n}$.

(b) $Es^T = e^{n\theta(s-1)}$, so $E(1-\frac{1}{n})^{2T} = \exp\{n\theta((1-\frac{1}{n})^2 - 1)\} = e^{-2\theta+\theta/n}$, and

$$\text{Var}\left((1-\tfrac{1}{n})^T\right) = e^{-2\theta}(e^{\theta/n} - 1) > e^{-2\theta}\frac{\theta}{n},$$

since $e^x - 1 > x$ for $x > 0$. No unbiased estimator attains the bound.

(c) By the delta method, $\sqrt{n}(e^{-\bar X} - e^{-\theta}) \xrightarrow{d} N(0, \theta e^{-2\theta})$. Also $n\,\text{Var}\left((1-\frac{1}{n})^T\right) = e^{-2\theta}n(e^{\theta/n}-1) \to \theta e^{-2\theta}$: the UMVUE and the MLE are asymptotically equivalent and asymptotically attain the bound.

Solution 28

(a) $l(\theta) = -\frac{n}{2}\log\theta - \frac{1}{2\theta}\sum(x_i-\theta)^2 + c$, and

$$\dot l(\theta) = -\frac{n}{2\theta} + \frac{\sum x_i^2}{2\theta^2} - \frac{n}{2} = \frac{n}{2\theta^2}(m_2 - \theta - \theta^2).$$

It is positive for $\theta < \hat\theta$ and negative after, where $\hat\theta$ is the positive root of $\theta^2 + \theta - m_2 = 0$.

(b) $I(\theta) = \frac{1}{\theta} + \frac{1}{2\theta^2} = \frac{2\theta+1}{2\theta^2}$ (mean part plus variance part of a normal), so $\sqrt{n}(\hat\theta - \theta) \xrightarrow{d} N\big(0, \frac{2\theta^2}{2\theta+1}\big)$. (Check by the delta method: $\text{Var}\,X^2 = 2\theta^2 + 4\theta^3$ and $g'(m_2) = 1/(2\theta+1)$.)

(c) $\bar X$ has variance θ/n and S^2 asymptotic variance $2\theta^2/n$. So $ARE(\bar X, \hat\theta) = \frac{2\theta}{2\theta+1}$ and $ARE(S^2, \hat\theta) = \frac{1}{2\theta+1}$, both below 1.

Solution 29

(a) Normal: $E_{\theta_0}\frac{1}{2}\{(X-\theta_1)^2 - (X-\theta_0)^2\} = \frac{1}{2}(\theta_1-\theta_0)^2$. Poisson: $\log\frac{f_0}{f_1} = x\log\frac{\theta_0}{\theta_1} - \theta_0 + \theta_1$, so $KL = \theta_0\log\frac{\theta_0}{\theta_1} - \theta_0 + \theta_1$.

(b) $-KL = E_0\log\frac{f_1}{f_0} \le E_0\big(\frac{f_1}{f_0} - 1\big) = \int_{f_0>0} f_1 - 1 \le 0$, with equality iff $f_1/f_0 = 1$ with probability one.

(c) By the LLN, $\frac{1}{n}l_n(\theta) \to E_{\theta_0}\log f(X;\theta)$ for each θ, and by (b) this limit is uniquely maximized at θ_0. The maximizer of $\frac{1}{n}l_n$, the MLE, should then approach θ_0 (rigorous under uniform convergence or concavity).

Solution 30

(a) $X_1 \mid T = t \sim B(t, 1/n)$, so $E(\mathbf{1}\{X_1 = 0\} \mid T) = (1-1/n)^T$, the UMVUE of $e^{-\theta}$ (T is complete sufficient).

(b) $E(X_1X_2 \mid T=t) = P(X_1=X_2=1 \mid T=t) = \binom{n-2}{t-2}/\binom{n}{t} = \frac{t(t-1)}{n(n-1)}$; the UMVUE of θ^2 is $\frac{T(T-1)}{n(n-1)}$.

(c) Given $X_{(n)}=y$, $X_1=y$ with probability $\frac{1}{n}$ and otherwise is $U(0,y)$. So $E(2X_1 \mid X_{(n)}) = 2X_{(n)}\big(\frac{1}{n}+\frac{n-1}{2n}\big) = \frac{n+1}{n}X_{(n)}$.

Solution 31

(a) With σ^2 known, $\bar X$ is complete sufficient for μ. S^2 and $X_{(n)}-X_{(1)}$ do not change when every X_i is shifted, so their laws do not depend on μ: they are ancillary, and Basu's theorem gives independence.

(b) $X_{(n)}-\bar X$ is also location invariant, hence independent of $\bar X$. So $E(X_{(n)} \mid \bar X) = \bar X + E(X_{(n)} - \bar X) = \bar X + \sigma E(Z_{(n)})$ with $Z_i \overset{\text{iid}}{\sim} N(0,1)$: $c_n = EZ_{(n)}$.

(c) $X_1-\bar X \sim N(0, \sigma^2\frac{n-1}{n})$ is independent of $\bar X$, so $E(X_1^2 \mid \bar X) = \bar X^2 + \sigma^2\frac{n-1}{n}$.

Solution 32

(a) The density is $\exp\{-T/(2\sigma^2) - \frac{n}{2}\log(2\pi\sigma^2)\}$ in $T=\sum(x_i-\mu)^2$, with $-1/(2\sigma^2)$ increasing in σ^2. The UMP test rejects for large T: $T \ge \sigma_0^2\chi^2_\alpha(n)$, since $T/\sigma_0^2 \sim \chi^2(n)$ at σ_0^2. Power: $\gamma(\sigma^2) = P\big(\chi^2(n) \ge \sigma_0^2\chi^2_\alpha(n)/\sigma^2\big)$.

(b) With $\hat\sigma^2 = \frac{1}{n}\sum(x_i-\bar x)^2$, $l(\sigma^2) = -\frac{n}{2}\log\sigma^2 - \frac{n\hat\sigma^2}{2\sigma^2}$ at $\mu=\bar x$, so

$$2\{l(\hat\sigma^2) - l(\sigma_0^2)\} = n\Big(\frac{\hat\sigma^2}{\sigma_0^2} - 1 - \log\frac{\hat\sigma^2}{\sigma_0^2}\Big) = n(v-1-\log v).$$

This is convex in v with minimum at $v=1$, so we reject for $v \le c_1$ or $v \ge c_2$, where $nv = (n-1)S^2/\sigma_0^2 \sim \chi^2(n-1)$ under H_0.

Solution 33

(a) Reject when $\bar X \ge 10 + 1.645\cdot 4/\sqrt{n}$; the power is $\gamma(\mu) = \Phi\big(\sqrt{n}(\mu-10)/4 - 1.645\big)$.

(b) $\sqrt{n}\cdot\frac{2}{4} - 1.645 \ge 1.282$ gives $n \ge \{2(1.645+1.282)\}^2 = 34.3$: $n=35$.

(c) $\gamma(11) = \Phi(\sqrt{35}/4 - 1.645) = \Phi(-0.166) = 0.434$.

(d) The MP test against $\mu_1 > 10$ rejects for large $\bar X$, and against $\mu_1 < 10$ for small $\bar X$ (Neyman–Pearson, with uniqueness of MP tests). No single test is MP against both.

Solution 34

(a) $L(\theta) = \theta^{-n}\mathbf{1}\{\theta \ge x_{(n)}\}$ and $\hat\theta = X_{(n)}$, so $\Lambda = L(\hat\theta)/L(\theta_0) = (\theta_0/X_{(n)})^n$ if $X_{(n)} \le \theta_0$ and ∞ otherwise. We reject when $X_{(n)} > \theta_0$ or $X_{(n)} \le c$; the size is $P_{\theta_0}(X_{(n)} \le c) = (c/\theta_0)^n = \alpha$, so $c = \theta_0\alpha^{1/n}$.

(b) Under H_0, $U = (X_{(n)}/\theta_0)^n \sim U(0,1)$, and $2\log\Lambda = -2\log U \sim \text{Exp}(2) = \chi^2(2)$. The usual $\chi^2(1)$ limit requires the regularity conditions; here the support depends on θ, and the MLE converges at rate n rather than $\sqrt{n}$.

Solution 35

(a) $\hat p = 0.6$.

- LRT: $2\{60\log 1.2 + 40\log 0.8\} = 2(10.939 - 8.926) = 4.03$.
- Wald: $\frac{(0.6-0.5)^2}{0.6\cdot 0.4/100} = 4.17$.
- Score: $\frac{(60-50)^2}{100\cdot 0.25} = 4.00$.

(b) Each is asymptotically $\chi^2(1)$ under H_0. All exceed 3.84, so H_0 is rejected (the score statistic only barely).

(c) The score statistic uses only p_0. The score confidence set is $\{p : (60-100p)^2 \le 3.84\cdot 100p(1-p)\}$, an interval found by solving the quadratic.

Solution 36

(a) $2n\bar{X}/\theta \sim \chi^2(2n)$ gives $\Big[\dfrac{2n\bar{X}}{\chi^2_{\alpha/2}(2n)}, \dfrac{2n\bar{X}}{\chi^2_{1-\alpha/2}(2n)}\Big]$.

(b) $W = X_{(n)}/\theta$ has cdf w^n, so the coverage is $P(a \le W \le b) = b^n - a^n = 1 - \alpha$, and the length is $X_{(n)}(1/a - 1/b)$. Treating $a = a(b)$ through $a^n = b^n - (1-\alpha)$, $a'(b) = b^{n-1}/a^{n-1}$, and

$$\frac{d}{db}\Big(\frac{1}{a} - \frac{1}{b}\Big) = -\frac{b^{n-1}}{a^{n+1}} + \frac{1}{b^2} < 0,$$

because $a < b$. The length decreases in b, so $b = 1$ and $a = \alpha^{1/n}$ are optimal.

(c) $\Big[\sqrt{(n-1)S^2/\chi^2_{\alpha/2}(n-1)},\ \sqrt{(n-1)S^2/\chi^2_{1-\alpha/2}(n-1)}\Big]$.

Solution 37

(a) $P(X_{n+1} = 1 \mid \mathbf{x}) = E(\theta \mid \mathbf{x}) = \frac{t+\alpha}{n+\alpha+\beta}$; with a uniform prior and $t = n$ this is $\frac{n+1}{n+2}$ (Laplace's rule of succession).

(b) The posterior variance $v = (n/\sigma^2 + 1/\tau^2)^{-1}$ does not depend on $\mathbf{x}$, so the Bayes risk of the posterior mean is $E\{\mathrm{Var}(\theta \mid \mathbf{X})\} = v < \sigma^2/n$. The Bayes risk of $\bar{X}$ is its constant MSE σ^2/n.

(c) The posterior mean is $w\bar{X} + (1-w)\mu_0$ with $w = \frac{n/\sigma^2}{n/\sigma^2 + 1/\tau^2} \in (0,1)$. Its MSE is $w^2\frac{\sigma^2}{n} + (1-w)^2(\theta-\mu_0)^2$, which exceeds $\frac{\sigma^2}{n}$ iff $(\theta-\mu_0)^2 > \frac{(1+w)\sigma^2}{(1-w)n}$.

Solution 38

(a) Under H_0 the rank vector is uniform over permutations, so each rank is uniform on $\{1,\dots,N\}$: mean $\frac{N+1}{2}$, variance $\frac{N^2-1}{12}$. All N ranks sum to a constant, so $0 = N\,\mathrm{Var}\,R + N(N-1)c$, giving $c = -\frac{N+1}{12}$.

(b) $EW = \frac{n(N+1)}{2}$ and

$$\mathrm{Var}\,W = n\frac{N^2-1}{12} - n(n-1)\frac{N+1}{12} = \frac{n(N+1)(N-n)}{12} = \frac{mn(N+1)}{12}.$$

(c) The $\binom{6}{3} = 20$ subsets of $\{1,\dots,6\}$ are equally likely; sums ≥ 14 come only from $\{4,5,6\}$ and $\{3,5,6\}$. So $P(W \ge 14) = 2/20 = 0.1$.

Solution 39

(a) Problem 1(a) with x/n gives $\frac{nx}{n+x} < n\log(1+x/n) < x$. Both bounds tend to x, and exp is continuous.

(b) $|x^k/k!| \le a^k/k!$ on $[-a,a]$, and $\sum a^k/k! < \infty$ by the ratio test; apply the Weierstrass M-test.

(c) The series of derivatives $\sum_{k\ge1} x^{k-1}/(k-1)!$ is the same series, uniformly convergent on $[-a,a]$, and the original converges at 0. By the theorem on term-by-term differentiation, $E'(x) = \sum_{k\ge1} x^{k-1}/(k-1)! = E(x)$.

(d) By the MVT, $f'' \ge 0$ makes f' nondecreasing. For $x < y < z$, the MVT gives $\frac{f(y)-f(x)}{y-x} = f'(\xi_1) \le f'(\xi_2) = \frac{f(z)-f(y)}{z-y}$, with $\xi_1 < y < \xi_2$. With $y = tx + (1-t)z$, this rearranges to $f(y) \le tf(x) + (1-t)f(z)$.

Solution 40

(a) If $P\mathbf{v} = \lambda\mathbf{v}$, $\mathbf{v} \ne \mathbf{0}$, then $\lambda\mathbf{v} = P^2\mathbf{v} = \lambda^2\mathbf{v}$, so $\lambda \in \{0,1\}$. A symmetric P is orthogonally diagonalizable, so its rank is the number of eigenvalues equal to 1, which is their sum, $\mathrm{tr}\,P$.

(b) $H' = H$, $H^2 = X(X'X)^{-1}X'X(X'X)^{-1}X' = H$, and $\mathrm{tr}\,H = \mathrm{tr}\{(X'X)^{-1}X'X\} = p$. The same holds for $I - H$, with trace $n - p$, and $(I-H)X = X - X = 0$.

(c) $(I+\mathbf{ab}')\big(I-\frac{\mathbf{ab}'}{1+\mathbf{b}'\mathbf{a}}\big)=I+\mathbf{ab}'\big(1-\frac{1+\mathbf{b}'\mathbf{a}}{1+\mathbf{b}'\mathbf{a}}\big)=I$. The matrix $\mathbf{ab}'$ has eigenvalue $\mathbf{b}'\mathbf{a}$ (eigenvector $\mathbf{a}$) and 0 with multiplicity $k-1$, so $\det(I+\mathbf{ab}')=1+\mathbf{b}'\mathbf{a}$.

(d) For $\rho\neq 1$, $\Sigma=(1-\rho)\big(I+\frac{\rho}{1-\rho}\mathbf{1}\mathbf{1}'\big)$, so

$$\Sigma^{-1}=\frac{1}{1-\rho}\Big(I-\frac{\rho}{1+(k-1)\rho}\mathbf{1}\mathbf{1}'\Big).$$

The eigenvalues of Σ are $1+(k-1)\rho$ (eigenvector $\mathbf{1}$) and $1-\rho$ (multiplicity $k-1$), so Σ is positive definite iff $-\frac{1}{k-1}<\rho<1$.

Solution 41

(a) The square $\{|x|+|y|<w\}$ has area $2w^2$, so $P(W_1\le w)=(w/\theta)^2$ on $(0,\theta)$. The joint pdf of the sample is $(2\theta^2)^{-n}\mathbf{1}\{M<\theta\}$, so M is sufficient by factorization.

(b) M has cdf $(m/\theta)^{2n}$ and pdf $2nm^{2n-1}/\theta^{2n}$. If $\int_0^\theta g(m)m^{2n-1}dm=0$ for all $\theta>0$, differentiating gives $g(\theta)\theta^{2n-1}=0$, so $g=0$ a.e.

(c) $EM=\frac{2n}{2n+1}\theta$ and $EM^2=\frac{2n}{2n+2}\theta^2$. By Lehmann–Scheffé the UMVUEs are $\frac{2n+1}{2n}M$ for θ and $\frac{2(n+1)}{n}M^2$ for $2\theta^2$.

Solution 42

(a) With $D=\theta_2-\theta_1$, $X_i=\theta_1+DU_i$ for $U_i\overset{\text{iid}}{\sim}U(0,1)$, so $EX_{(1)}=\theta_1+\frac{D}{n+1}$ and $EX_{(n)}=\theta_2-\frac{D}{n+1}$.

(b) $E(X_{(n)}-X_{(1)})=\frac{n-1}{n+1}D$, so $\frac{n+1}{n-1}(X_{(n)}-X_{(1)})$ is the UMVUE of D. The midpoint has UMVUE $\frac{1}{2}(X_{(1)}+X_{(n)})$. For θ_1: $X_{(1)}-\frac{X_{(n)}-X_{(1)}}{n-1}$, whose mean is $\theta_1+\frac{D}{n+1}-\frac{D}{n+1}=\theta_1$. All are functions of the complete sufficient statistic.

Solution 43

(a) $p_0=\mathbf{1}\{0<x_{(1)},x_{(n)}<1\}$ and $p_{\theta_1}=\mathbf{1}\{\theta_1<x_{(1)},x_{(n)}<\theta_1+1\}$. The ratio is ∞ where $x_{(n)}\ge 1$ (only $p_{\theta_1}>0$), equals 1 where $\theta_1<x_{(1)}$ and $x_{(n)}<1$, and equals 0 where $x_{(1)}\le\theta_1$.

(b) Size: $P_0(X_{(1)}\ge c)=(1-c)^n=\alpha$, and $P_0(X_{(n)}\ge 1)=0$. Fix θ_1. If $\theta_1<c$, the test rejects everywhere the ratio is ∞, never where it is 0, and on part of the region where it equals $k=1$. This is the Neyman–Pearson form with $k=1$ and size α, so the test is MP. If $\theta_1\ge c$, its power at θ_1 is 1, which cannot be beaten. The test does not depend on θ_1, so it is UMP.

(c) For $0<\theta<c$, $1-P_\theta(X_{(n)}<1,X_{(1)}<c)=1-\{(1-\theta)^n-(1-c)^n\}=1-(1-\theta)^n+\alpha$. For $\theta\ge c$ the power is 1.

Solution 44

For $\theta_1>\theta_0$, $p_{\theta_1}/p_{\theta_0}=(\theta_0/\theta_1)^n=:k$ on $\{x_{(n)}\le\theta_0\}$ and ∞ on $\{\theta_0<x_{(n)}\le\theta_1\}$. Any test that rejects on $\{x_{(n)}>\theta_0\}$ and has size α at θ_0 is of Neyman–Pearson form with this k, hence MP. Both tests reject when $x_{(n)}>\theta_0$. At θ_0, $E\phi_1=1-(1-\alpha)=\alpha$ and $E\phi_2=0+\alpha=\alpha$. Their powers for $\theta>\theta_0$ are

$$1-(1-\alpha)\Big(\frac{\theta_0}{\theta}\Big)^n\quad\text{and}\quad 1-\Big(\frac{\theta_0}{\theta}\Big)^n+\alpha\Big(\frac{\theta_0}{\theta}\Big)^n,$$

which are equal. For $\theta\le\theta_0$ both powers are at most α: for ϕ_2 it is exactly α, and for ϕ_1 it is $\max\{0,1-(1-\alpha)(\theta_0/\theta)^n\}\le\alpha$. So both are UMP for the composite null.

Solution 45

(a) $l(\mu)=-\frac{n}{2}(\bar x-\mu)^2+\text{const}$ is maximized over $\mu\ge 0$ at $\hat\mu=\bar X^+$. So $2\log\Lambda=2\{l(\hat\mu)-l(0)\}=n\{\bar X^2-(\bar X-\bar X^+)^2\}=n(\bar X^+)^2$.

(b) Under H_0, $Z=\sqrt{n}\bar X\sim N(0,1)$ and $2\log\Lambda=(Z^+)^2$, which is 0 with probability $\frac{1}{2}$ and otherwise distributed as Z^2 given $Z>0$, i.e. $\chi^2(1)$. Hence $P(2\log\Lambda>c)=\frac{1}{2}P(\chi^2(1)>c)=\alpha$

gives $c = \chi^2_{2\alpha}(1) = z^2_\alpha$: the test rejects when $\sqrt{n}\bar{X} > z_\alpha$. The naive critical value $\chi^2_\alpha(1)$ gives actual size $\alpha/2$.

(c) For $\mu > 0$, $P(\bar{X} < 0) \to 0$, so $\sqrt{n}(\hat{\mu} - \mu) \xrightarrow{d} N(0,1)$. For $\mu = 0$, $\sqrt{n}\hat{\mu} = (\sqrt{n}\bar{X})^+ \sim Z^+$ exactly: a point mass $\frac{1}{2}$ at 0 plus the positive half of $N(0,1)$. The MLE is not asymptotically normal at the boundary.

Solution 46

(a) For $x > -\log n$, $P(X_{(n)} \le \log n + x) = (1 - e^{-x}/n)^n \to \exp(-e^{-x})$.

(b) $P(nX_{(1)} > s) = (e^{-s/n})^n = e^{-s}$: Exp(1) exactly.

(c) For $s > 0$ and $x > -\log n$,

$$P\big(nX_{(1)} > s,\ X_{(n)} - \log n \le x\big) = P\big(s/n < X_i \le \log n + x\ \forall i\big) = \Big(e^{-s/n} - \frac{e^{-x}}{n}\Big)^n \to e^{-s}\exp(-e^{-x}),$$

since $n\{e^{-s/n} - 1 - e^{-x}/n\} \to -s - e^{-x}$. The joint limit cdf factors, so the limits are independent.

Solution 47

(a) An uncensored observation contributes $\theta^{-1}e^{-z/\theta}$ and a censored one $P(X > c) = e^{-c/\theta}$. Hence $L(\theta) = \theta^{-D}e^{-T/\theta}$, and (D, T) is sufficient.

(b) $l = -D\log\theta - T/\theta$, so $\dot{l} = -D/\theta + T/\theta^2$, giving $\hat{\theta} = T/D$ (total time on test divided by the number of failures).

(c) $ED = nP_\theta$ and $E\min(X, c) = \int_0^c P(X > x)dx = \theta P_\theta$. With $\ddot{l} = D/\theta^2 - 2T/\theta^3$, $I_n(\theta) = -E\ddot{l} = -\frac{nP_\theta}{\theta^2} + \frac{2n\theta P_\theta}{\theta^3} = \frac{nP_\theta}{\theta^2}$. So $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \theta^2/P_\theta)$. Without censoring the variance would be θ^2; censoring inflates it by the factor $1/P_\theta$, the inverse of the probability of observing a failure.

Part II. Past exam problems (restated)

The following are the mathematical statistics problems of the SNU Statistics graduate entrance exams from October 2020 to April 2026, restated in our own words with the original mathematical content. Regression-only problems are omitted. Tags give the exam date and problem number; the original papers are on the department website.

Past problem *[2020-10 #1]*

$X_1, \dots, X_n \overset{\text{iid}}{\sim} N(0,1)$ with order statistics $X_{(1)} < \cdots < X_{(n)}$, $X_{(0)} = -\infty$, $X_{(n+1)} = \infty$. Let

$$T_n = \sum_{i=k+1}^{n} i \int_{X_{(n-i)}}^{X_{(n-i+1)}} \frac{\phi(x)}{1-\Phi(x)} dx, \qquad k < n.$$

(a) Find the distribution of T_n. (b) For $n = 2k$ and $k \to \infty$, find a_k, b_k such that the mgf of $a_k(T_n - b_k)$ converges to that of $N(0,1)$, and prove it.

Past problem *[2020-10 #2]*

$X_i \sim N(\mu_i, 1)$, $i = 1, 2$, independent, with parameter space $\Omega = \{\mu_1 \geq 0, \mu_2 \geq 0\}$. Derive the LRT of $H_0 : (\mu_1, \mu_2) = (0,0)$ vs $H_1 : (\mu_1, \mu_2) \neq (0,0)$ and the condition determining its rejection region.

Past problem *[2020-10 #3]*

For a sample from $N(\mu, \sigma^2)$ with σ^2 known, find the UMVUE of $e^{c\mu}$ ($c \neq 0$ known), and decide whether it attains the Cramér–Rao bound.

Past problem *[2021-04 #3]*

$X_1, \dots, X_n$ have pdf $\theta^{-1} e^{-x/\theta}$, $x > 0$, where the parameter space is $\theta \geq 1$. Find the MLE and the limit law of $\sqrt{n}(\hat{\theta} - \theta)$.

Past problem *[2021-04 #4]*

$X_1, \dots, X_n \overset{\text{iid}}{\sim} U[0, \theta]$. Find a UMP test of $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$.

Past problem *[2021-04 #5]*

(X_i, Y_i), $i \leq n$, are i.i.d. uniform on the triangle $\{x > 0, y > 0, x + y < \theta\}$. Find the UMVUE of θ and its pdf.

Past problem *[2021-10 #3]*

$X_1, \dots, X_n$ have pdf $\sigma^{-1} e^{-(x-\mu)/\sigma}$, $x \geq \mu$, with both parameters unknown. Find the LRT of $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$.

Past problem *[2021-10 #4]*

(X_i, Y_i) are i.i.d. uniform in the disk of radius r centred at the origin, and $\theta = r^2$. Find the UMVUE of θ and its pdf.

Past problem *[2021-10 #5]*

$X_1, \dots, X_n \overset{\text{iid}}{\sim} N(0,1)$. Prove or disprove: $n\Phi(\min X_i)$ and $n\{1 - \Phi(\max X_i)\}$ are asymptotically independent.

Past problem *[2022-04 #1]*

$X_1, \dots, X_m \overset{\text{iid}}{\sim} U(0, \theta)$ and, independently, $Y_1, \dots, Y_n \overset{\text{iid}}{\sim} U(0, \eta)$. Find the UMVUE of θ/η.

Past problem *[2022-04 #2]*

A single X has pdf $e^{x-\theta}/(1 + e^{x-\theta})^2$. Show that the UMP level-α test of $H_0 : \theta \leq 0$ vs $H_1 : \theta > 0$ rejects when $X > C$, and find C.

Past problem *[2022-04 #3]*

Buses arrive at a stop as a Poisson process of rate 1 per hour, and each stays 10 minutes. Given that two buses arrived between noon and 1 pm, find the probability that both were at the stop at the same time.

Past problem *[2022-10 #1]*

$X_1, \ldots, X_n$ have the logistic pdf $e^x/(1+e^x)^2$. Find the pdf of $\log \dfrac{1+e^{-X_{(1)}}}{1+e^{-X_{(n)}}}$.

Past problem *[2022-10 #2]*

$\mathrm{Corr}(X_n, Y_n) \to 1$, and $(X_n - EX_n)/\sqrt{\mathrm{Var}\, X_n} \xrightarrow{d} X$. Show that $(Y_n - EY_n)/\sqrt{\mathrm{Var}\, Y_n} \xrightarrow{d} X$.

Past problem *[2022-10 #4]*

$X_1, \ldots, X_n$ have the Pareto pdf $\theta a^\theta x^{-(\theta+1)}$, $x > a$, with $\theta > 0$ known. Find a level-α UMP test of $H_0 : a = a_0$ vs $H_1 : a \neq a_0$.

Past problem *[2022-10 #5]*

$\mathbf{y} \sim N_n(\mu, V)$ with V positive definite, and A, B symmetric idempotent. Show that (1) $\mathbf{y}'A\mathbf{y}$ and $B\mathbf{y}$ are independent iff $BVA = 0$; (2) $\mathbf{y}'A\mathbf{y}$ and $\mathbf{y}'B\mathbf{y}$ are independent iff $AVB = 0$.

Past problem *[2022-10 #6]*

$\mathbf{y} \sim N_n(\mathbf{0}, I)$ and X is $n \times k$ of rank $k < n$, $H = X(X'X)^{-1}X'$. Find the distributions of $\mathbf{y}'H\mathbf{y}$, $\mathbf{y}'(I-H)\mathbf{y}$ and $T = \mathbf{y}'H\mathbf{y}/\mathbf{y}'(I-H)\mathbf{y}$.

Past problem *[2023-04 #1]*

For a random vector $\mathbf{y}$ with mean μ and covariance Σ and symmetric A, show $E(\mathbf{y}'A\mathbf{y}) = \mathrm{tr}(A\Sigma) + \mu'A\mu$.

Past problem *[2023-04 #3]*

$U_1, U_2, \ldots \overset{\text{iid}}{\sim} U(0,1)$, independent of X with $P(X = x) = \frac{c}{x!}$, $x = 1, 2, \ldots$, $c = 1/(e-1)$. For $Z = \min\{U_1, \ldots, U_X\}$, compute $P(Z \le z)$, $0 < z < 1$.

Past problem *[2023-04 #4]*

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$, $n > 1$. Show that $u(X_1) = \mathbf{1}\{X_1 \le c\}$ is unbiased for $\Phi(c-\theta)$, and use it to find the UMVUE of $\Phi(c-\theta)$.

Past problem *[2023-04 #5]*

$X_1, \ldots, X_n$ have the Weibull pdf $\lambda c x^{c-1} e^{-\lambda x^c}$, $x > 0$, c known. (1) Show that X_i^c is exponential. (2) Find the UMP level-α test of $H_0 : \lambda \ge \lambda_0$ vs $H_1 : \lambda < \lambda_0$.

Past problem *[2023-10 #1]*

For a Poisson(λ) sample, estimate $\theta = e^{-\lambda}$ by $\hat\theta^{(1)} = e^{-\bar X}$ or $\hat\theta^{(2)} = \frac{1}{n}\sum \mathbf{1}\{X_i = 0\}$. Find both limit laws and decide which is more efficient.

Past problem *[2023-10 #2]*

ϕ is the UMP size-α test of $H_0 : f = f_0$ vs $H_1 : f = f_1$, with power $\beta \in (0,1)$. Find the UMP size-$(1-\beta)$ test of $H_0^* : f = f_1$ vs $H_1^* : f = f_0$ in terms of ϕ.

Past problem *[2023-10 #3]*

$X_1, \ldots, X_n$ are i.i.d. uniform on $(\theta, \theta + |\theta|)$, $\theta \neq 0$. Find the MME and the MLE of θ, and decide whether the MLE is consistent.

Past problem *[2024-05 #1]*

For a normal sample, use $(n-1)S_n^2/\sigma^2 \sim \chi^2(n-1)$ to show $\sqrt{n-1}(S_n-\sigma) \xrightarrow{d} N(0,\sigma^2/2)$.

Past problem *[2024-05 #2]*

X_i are i.i.d. with $P(X_i=\sigma)=P(X_i=-\sigma)=\frac{1}{2}$. Show that $S_n-\sigma=o_p\big((n-1)^{-\gamma}\big)$ for every $\gamma<1$.

Past problem *[2024-05 #3]*

For a sample from $U(\theta,\theta+1)$, $n\geq 2$, find a minimal sufficient statistic.

Past problem *[2024-05 #4]*

$\mathbf{y}\sim N(X\beta,\sigma^2 I_n)$, with X of size $n\times(p+1)$ and full column rank, and H the hat matrix. Find the distribution of $F=\mathbf{y}'H\mathbf{y}/\mathbf{y}'(I-H)\mathbf{y}$.

Past problem *[2024-10 #1]*

For $U(0,\theta)$ and $H_0:\theta=\theta_0$ vs $H_1:\theta=\theta_1$ $(\theta_0<\theta_1)$, show that any test with $\phi=1$ when $x_{(n)}>\theta_0$ and $E_{\theta_0}\phi=\alpha$ is most powerful.

Past problem *[2024-10 #2]*

If $X_n\xrightarrow{d}Z$, where all cdfs are continuous, show $X_n=O_p(1)$.

Past problem *[2024-10 #3]*

Show that a compact subset of a metric space is complete.

Past problem *[2025-04 #1]*

Let $X_n,Y_n=O_p(1)$ and $Z_n=o_p(1)$. Prove (1) $X_n+Y_n=O_p(1)$; (2) $X_nY_n=O_p(1)$; (3) $Z_nX_n=o_p(1)$; (4) $Z_n=O_p(1)$.

Past problem *[2025-04 #2]*

For a Beta$(\alpha,1)$ sample: (1) show $\hat{\alpha}^{\mathrm{MME}}=\bar{X}/(1-\bar{X})$ and find its limit law; (2) find the MLE and its limit law; (3) compute the ARE of the MME with respect to the MLE.

Past problem *[2025-04 #3]*

For the signed rank statistic $W_n=\sum \mathrm{sgn}(X_i-\theta_0)R(|X_i-\theta_0|)$ under $H_0:\theta=\theta_0$ (symmetric continuous f): (1) show $W_n\overset{d}{=}\sum_j jS_j$ with i.i.d. fair signs S_j; (2) show $W_n/\sqrt{\mathrm{Var}\,W_n}\xrightarrow{d}N(0,1)$; (3) show $W_n^+\overset{d}{=}\sum_j jB_j$ with $B_j\overset{\mathrm{iid}}{\sim}$ Bernoulli$(\frac{1}{2})$.

Past problem *[2025-10 #1]*

$X_1,\dots,X_n$ have pdf $2x/\theta^2$ on $(0,\theta)$. (a) Find the MLE and the MME. (b) Find their means and variances. (c) Compare them by MSE.

Past problem *[2025-10 #2]*

$(X_1,X_2,X_3)'\sim N_3(\mathbf{0},\Sigma)$ with $\Sigma=\begin{pmatrix}1&\rho&\rho^2\\ \rho&1&\rho\\ \rho^2&\rho&1\end{pmatrix}$, $|\rho|<1$. (a) Find the joint mgf. (b) Find $\mathrm{Var}(X_1-2X_2+X_3)$. (c) Are X_1-X_2 and X_2-X_3 independent?

Past problem *[2025-10 #3]*

(a) State and prove the mean value theorem. (b) Give a function satisfying its hypotheses and explain. (c) Give a continuous function for which the conclusion fails, and explain why.

Past problem *[2026-04 #1]*

$(X_1, X_2, X_3) \sim \text{Multi}(n; p_1, p_2, p_3)$, reparametrized by $\theta_1 = p_1$, $\theta_2 = p_2/(p_2 + p_3)$. (a) Find the MLE. (b) Find the Fisher information matrix. (c) Find the limit law of $(\hat{\theta}_1, \hat{\theta}_2)$.

Past problem *[2026-04 #2]*

(X_1, X_2, X_3) is uniform on the unit ball. (a) Find the joint pdf of (X_1, X_2). (b) Find the marginal pdf of X_1. (c) Find the pdf of X_1 given $X_2 = 0$. (d) Find $\text{Cov}(\mathbf{X})$.

Past problem *[2026-04 #3]*

$f_n(x) = nx(1-x)^n$ on $[0, 1]$. (a) Find the pointwise limit. (b) Is the convergence uniform?

Part II solutions

Solution 2020-10 #1

Since $\frac{d}{dx}\{-\log(1-\Phi(x))\} = \frac{\phi(x)}{1-\Phi(x)}$, put $E_i = -\log(1-\Phi(X_i))$. These are i.i.d. Exp(1) (because $1-\Phi(X_i) \sim U(0,1)$), and the map is increasing, so $E_{(j)} = -\log(1-\Phi(X_{(j)}))$ with $E_{(0)} = 0$. The integral is $E_{(n-i+1)} - E_{(n-i)}$. With $j = n-i+1$, the term is $(n-j+1)(E_{(j)} - E_{(j-1)}) =: D_j$, and the normalized spacings $D_1, \dots, D_n$ are i.i.d. Exp(1).

(a) $T_n = \sum_{j=1}^{n-k} D_j \sim \text{Gamma}(n-k, 1)$.

(b) For $n = 2k$, $T_n \sim \text{Gamma}(k, 1)$. Take $a_k = k^{-1/2}$, $b_k = k$. The cgf of $(T_n - k)/\sqrt{k}$ is

$$-t\sqrt{k} - k\log\Big(1 - \frac{t}{\sqrt{k}}\Big) = -t\sqrt{k} + k\Big(\frac{t}{\sqrt{k}} + \frac{t^2}{2k} + O(k^{-3/2})\Big) = \frac{t^2}{2} + O(k^{-1/2})$$

for $|t| < \sqrt{k}$, so the mgf converges to $e^{t^2/2}$.

Solution 2020-10 #2

The MLE over Ω is $\hat{\mu}_i = X_i^+$, so

$$2\log\Lambda = \sum_i \{X_i^2 - (X_i - X_i^+)^2\} = (X_1^+)^2 + (X_2^+)^2.$$

Reject when this is at least c. Under H_0, each $(X_i^+)^2$ is 0 or $\chi^2(1)$ with probability $\frac{1}{2}$ each, and the number of positive X_i is $B(2, \frac{1}{2})$. Hence the null law is $\frac{1}{4}\delta_0 + \frac{1}{2}\chi^2(1) + \frac{1}{4}\chi^2(2)$, and c solves $\frac{1}{2}P(\chi^2(1) \ge c) + \frac{1}{4}P(\chi^2(2) \ge c) = \alpha$.

Solution 2020-10 #3

$\bar{X} \sim N(\mu, \sigma^2/n)$ is complete sufficient, and $Ee^{c\bar{X}} = e^{c\mu + c^2\sigma^2/(2n)}$. So the UMVUE is $e^{c\bar{X} - c^2\sigma^2/(2n)}$, with variance $e^{2c\mu}(e^{c^2\sigma^2/n} - 1)$. The CR bound is $\frac{(ce^{c\mu})^2}{n/\sigma^2} = e^{2c\mu}\frac{c^2\sigma^2}{n}$. Since $e^x - 1 > x$ for $x > 0$, the bound is not attained.

Solution 2021-04 #3

$l(\theta) = -n\log\theta - n\bar{x}/\theta$ increases on $(0, \bar{x})$ and decreases afterwards, so on $[1, \infty)$ the MLE is $\hat{\theta} = \max(\bar{X}, 1)$. If $\theta > 1$, $P(\bar{X} < 1) \to 0$, so $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \theta^2)$. If $\theta = 1$, $\sqrt{n}(\hat{\theta} - 1) = \{\sqrt{n}(\bar{X} - 1)\}^+ \xrightarrow{d} Z^+$, $Z \sim N(0,1)$: half the mass sits at 0.

Solution 2021-04 #4

Reject when $X_{(n)} > \theta_0$ or $X_{(n)} \le \theta_0\alpha^{1/n}$; the size is $0 + \alpha$. For $\theta_1 > \theta_0$ the test is MP by Problem 44 (it rejects on $\{x_{(n)} > \theta_0\}$ and has size α). For $\theta_1 < \theta_0$ the ratio $p_{\theta_1}/p_{\theta_0}$ is $(\theta_0/\theta_1)^n = k$ on $\{x_{(n)} \le \theta_1\}$ and 0 on $(\theta_1, \theta_0]$. If $\theta_0\alpha^{1/n} \le \theta_1$, the rejection region lies where the ratio equals k, so the test is of NP form and MP. Otherwise its power at θ_1 is $P_{\theta_1}(X_{(n)} \le \theta_0\alpha^{1/n}) = 1$. Hence it is UMP.

Solution 2021-04 #5

$W = X + Y$ has $P(W \le w) = (w/\theta)^2$. The joint pdf is $(2/\theta^2)^n \mathbf{1}\{M < \theta\}$ with $M = \max W_i$, which is sufficient and (as in Problem 41) complete, with cdf $(m/\theta)^{2n}$ and $EM = \frac{2n}{2n+1}\theta$. The UMVUE is $T = \frac{2n+1}{2n}M$, with pdf

$$f_T(t) = \frac{2n\,t^{2n-1}}{(c\theta)^{2n}}, \qquad 0 < t < c\theta, \quad c = \frac{2n+1}{2n}.$$

Solution 2021-10 #3

Unrestricted: $\hat\mu = X_{(1)}$ and $\hat\sigma = \bar X - X_{(1)}$. Under H_0 (possible only if $X_{(1)} \ge \mu_0$): $\hat\sigma_0 = \bar X - \mu_0 = \hat\sigma + (X_{(1)} - \mu_0)$. In both cases the exponential factor equals e^{-n}, so $\Lambda = (\hat\sigma_0/\hat\sigma)^n = \{1 + (X_{(1)} - \mu_0)/\hat\sigma\}^n$. Reject when $X_{(1)} < \mu_0$ or $(X_{(1)} - \mu_0)/\hat\sigma$ is large. Under H_0, $2n(X_{(1)} - \mu_0)/\sigma \sim \chi^2(2)$ independently of $2n\hat\sigma/\sigma \sim \chi^2(2n-2)$ (Problem 26). So the test rejects when $X_{(1)} < \mu_0$ or

$$\frac{(n-1)(X_{(1)} - \mu_0)}{\hat\sigma} \ge F_\alpha(2, 2n-2).$$

Solution 2021-10 #4

$R_i^2 = X_i^2 + Y_i^2$ has $P(R^2 \le s) = \pi s/(\pi r^2) = s/\theta$, so $R_i^2 \overset{\text{iid}}{\sim} U(0,\theta)$, and $M = \max R_i^2$ is complete sufficient. The UMVUE is $T = \frac{n+1}{n} M$, with pdf $f_T(t) = nt^{n-1}/(c\theta)^n$ on $(0, c\theta)$, $c = \frac{n+1}{n}$.

Solution 2021-10 #5

True. With $U_i = \Phi(X_i) \overset{\text{iid}}{\sim} U(0,1)$, the two variables are $nU_{(1)}$ and $n(1 - U_{(n)})$. For $s, t \ge 0$ and $s + t < n$,

$$P\big(nU_{(1)} > s,\ n(1-U_{(n)}) > t\big) = P\Big(\frac{s}{n} < U_i < 1 - \frac{t}{n}\ \forall i\Big) = \Big(1 - \frac{s+t}{n}\Big)^n \to e^{-s}e^{-t}.$$

The joint survival function converges to that of two independent Exp(1) variables, so they are asymptotically independent.

Solution 2022-04 #1

$(X_{(m)}, Y_{(n)})$ is complete sufficient (product of independent complete families). $EX_{(m)} = \frac{m}{m+1}\theta$, and for $n \ge 2$, $EY_{(n)}^{-1} = \int_0^\eta y^{-1} \frac{n y^{n-1}}{\eta^n} dy = \frac{n}{(n-1)\eta}$. By independence the UMVUE is

$$\frac{m+1}{m} \cdot \frac{n-1}{n} \cdot \frac{X_{(m)}}{Y_{(n)}}.$$

Solution 2022-04 #2

For $\theta_0 < \theta_1$,

$$\frac{f(x;\theta_1)}{f(x;\theta_0)} = e^{-(\theta_1 - \theta_0)} \Big(\frac{1 + e^{x-\theta_0}}{1 + e^{x-\theta_1}}\Big)^2.$$

The derivative of the bracket has the sign of $e^{x-\theta_0} - e^{x-\theta_1} > 0$, so the family has monotone likelihood ratio in x. By Neyman–Pearson the MP test of θ_0 vs θ_1 rejects for $X > C$. Its power is increasing in θ (location family), so the test is UMP for $\theta \le 0$ vs $\theta > 0$ (as in Theorem 9.3 of the notes). Size: $P_0(X > C) = 1/(1 + e^C) = \alpha$, so $C = \log\frac{1-\alpha}{\alpha}$.

Solution 2022-04 #3

Given two arrivals in an hour, their times are i.i.d. $U(0, 60)$ minutes. They overlap iff $|U_1 - U_2| < 10$, which has probability $1 - (50/60)^2 = \frac{11}{36}$.

Solution 2022-10 #1

With $F(x) = 1/(1 + e^{-x})$, $1 + e^{-x} = 1/F(x)$, so the variable equals $\log\{F(X_{(n)})/F(X_{(1)})\}$. Let $V_i = -\log F(X_i) \overset{\text{iid}}{\sim} \text{Exp}(1)$; the map is decreasing, so the variable is $\max V_i - \min V_i$. By memorylessness this range has the law of the maximum of $n-1$ i.i.d. Exp(1) variables:

$$f(y) = (n-1)e^{-y}(1 - e^{-y})^{n-2}, \qquad y > 0.$$

Solution 2022-10 #2

Let A_n, B_n be the standardized X_n, Y_n. Then $E(A_n - B_n)^2 = 2 - 2\,\mathrm{Corr}(X_n, Y_n) \to 0$, so $B_n - A_n \xrightarrow{p} 0$ by Markov's inequality. Since $B_n = A_n + (B_n - A_n)$, Slutsky's theorem gives $B_n \xrightarrow{d} X$.

Solution 2022-10 #4

$X_{(1)}$ is sufficient with $P(X_{(1)} > t) = (a/t)^{n\theta}$, $t > a$. Reject when $X_{(1)} < a_0$ or $X_{(1)} \geq a_0 \alpha^{-1/(n\theta)}$; the size is $0 + (a_0/(a_0\alpha^{-1/(n\theta)}))^{n\theta} = \alpha$. The argument is that of Past problem 2021-04 #4 applied to $1/X_i$, a scale family on $(0, 1/a)$. For $a_1 > a_0$ the ratio is 0 on $(a_0, a_1]$ and constant on $x_{(1)} > a_1$, and the test is MP or has power 1. For $a_1 < a_0$ the ratio is ∞ on $(a_1, a_0]$, where the test rejects, and constant elsewhere. So the test is UMP.

Solution 2022-10 #5

Sufficiency. $A\mathbf{y}$ and $B\mathbf{y}$ are jointly normal with $\mathrm{Cov}(B\mathbf{y}, A\mathbf{y}) = BVA$. If $BVA = 0$ (equivalently $AVB = 0$), they are independent. Then $\mathbf{y}'A\mathbf{y} = \|A\mathbf{y}\|^2$ is a function of $A\mathbf{y}$, which gives both (1) and (2).

Necessity (sketch, $\mu = 0$). Write $\mathbf{y} = V^{1/2}\mathbf{z}$. In (1), independence gives $\mathrm{Cov}(\mathbf{y}'A\mathbf{y}, (\mathbf{b}'B\mathbf{y})^2) = 0$ for every $\mathbf{b}$, and this covariance equals $2\,\mathbf{b}'BVAVB\mathbf{b} = 2\|AVB\mathbf{b}\|^2$ (using $A = A^2$). Hence $AVB = 0$. In (2), $\mathrm{Cov}(\mathbf{y}'A\mathbf{y}, \mathbf{y}'B\mathbf{y}) = 2\,\mathrm{tr}(AVBV) = 2\,\mathrm{tr}(PQ)$ with $P = V^{1/2}AV^{1/2}$ and $Q = V^{1/2}BV^{1/2}$ nonnegative definite. $\mathrm{tr}(PQ) = \|P^{1/2}Q^{1/2}\|_F^2 = 0$ forces $PQ = 0$, i.e. $AVB = 0$. For general μ the same conclusion follows from the cumulants.

Solution 2022-10 #6

H and $I - H$ are symmetric idempotent with ranks k and $n-k$, and $H(I-H) = 0$. So $\mathbf{y}'H\mathbf{y} \sim \chi^2(k)$ and $\mathbf{y}'(I-H)\mathbf{y} \sim \chi^2(n-k)$, independently. Hence $\frac{n-k}{k}T \sim F(k, n-k)$, and $T/(1+T) \sim \mathrm{Beta}(\frac{k}{2}, \frac{n-k}{2})$.

Solution 2023-04 #1

$\mathbf{y}'A\mathbf{y} = \mathrm{tr}(A\mathbf{y}\mathbf{y}')$ and $E\mathbf{y}\mathbf{y}' = \Sigma + \mu\mu'$, so $E\mathbf{y}'A\mathbf{y} = \mathrm{tr}(A\Sigma) + \mathrm{tr}(A\mu\mu') = \mathrm{tr}(A\Sigma) + \mu'A\mu$.

Solution 2023-04 #3

$P(Z > z \mid X = x) = (1-z)^x$, so

$$P(Z > z) = c\sum_{x \geq 1} \frac{(1-z)^x}{x!} = \frac{e^{1-z} - 1}{e - 1}, \qquad P(Z \leq z) = \frac{e - e^{1-z}}{e-1}.$$

Solution 2023-04 #4

$Eu(X_1) = P(X_1 \leq c) = \Phi(c - \theta)$. $\bar{X}$ is complete sufficient, and $X_1 - \bar{X} \sim N(0, 1 - \frac{1}{n})$ is independent of $\bar{X}$, so

$$E\{u(X_1) \mid \bar{X}\} = P(X_1 - \bar{X} \leq c - \bar{X} \mid \bar{X}) = \Phi\Big(\frac{c - \bar{X}}{\sqrt{1 - 1/n}}\Big),$$

the UMVUE by Lehmann–Scheffé.

Solution 2023-04 #5

(1) $P(X^c > t) = P(X > t^{1/c}) = e^{-\lambda t}$: X^c is exponential with rate λ. (2) The joint pdf is $\lambda^n \exp(-\lambda T) \prod c x_i^{c-1}$ with $T = \sum X_i^c$, an exponential family in which small λ makes large T likely. The UMP test of $\lambda \geq \lambda_0$ vs $\lambda < \lambda_0$ rejects for large T. Since $2\lambda_0 T \sim \chi^2(2n)$ at λ_0, it rejects when $2\lambda_0 \sum X_i^c \geq \chi^2_\alpha(2n)$.

Solution 2023-10 #1

(1) By the delta method, $\sqrt{n}(\hat{\theta}^{(1)} - \theta) \xrightarrow{d} N(0, \lambda e^{-2\lambda})$. (2) $\hat{\theta}^{(2)}$ is a mean of i.i.d. Bernoulli$(e^{-\lambda})$,

so $\sqrt{n}(\hat{\theta}^{(2)} - \theta) \xrightarrow{d} N(0, e^{-\lambda}(1 - e^{-\lambda}))$. (3) $ARE(\hat{\theta}^{(2)}, \hat{\theta}^{(1)}) = \frac{\lambda e^{-2\lambda}}{e^{-\lambda}(1-e^{-\lambda})} = \frac{\lambda}{e^{\lambda}-1} < 1$, so $\hat{\theta}^{(1)}$ (the MLE) is more efficient.

Solution 2023-10 #2

The UMP test has NP form: $\phi = 1$ where $f_1 > kf_0$ and $\phi = 0$ where $f_1 < kf_0$, for some $k > 0$. Then $1 - \phi = 1$ where $f_0 > k^{-1}f_1$ and 0 where $f_0 < k^{-1}f_1$: NP form for f_1 vs f_0. Its size under f_1 is $1 - \beta$ and its power under f_0 is $1 - \alpha$. So $\phi^* = 1 - \phi$ is the UMP size-$(1 - \beta)$ test.

Solution 2023-10 #3

For $\theta > 0$ the support is $(\theta, 2\theta)$, and for $\theta < 0$ it is $(\theta, 0)$; the sign of the data reveals the sign of θ. (1) $EX = \frac{3}{2}\theta$ for $\theta > 0$ and $\frac{1}{2}\theta$ for $\theta < 0$, so $\tilde{\theta} = \frac{2}{3}\bar{X}$ if $\bar{X} > 0$ and $2\bar{X}$ if $\bar{X} < 0$. (2) For positive data, $L = \theta^{-n}\mathbf{1}\{x_{(n)}/2 \le \theta \le x_{(1)}\}$ gives $\hat{\theta} = X_{(n)}/2$. For negative data, $L = |\theta|^{-n}\mathbf{1}\{\theta \le x_{(1)}\}$ gives $\hat{\theta} = X_{(1)}$ (with closed support). (3) Yes: $X_{(n)} \xrightarrow{p} 2\theta$ for $\theta > 0$ and $X_{(1)} \xrightarrow{p} \theta$ for $\theta < 0$.

Solution 2024-05 #1

$(n - 1)S_n^2/\sigma^2$ is distributed as a sum of $n - 1$ i.i.d. $\chi^2(1)$ variables (mean 1, variance 2), so by the CLT $\sqrt{n-1}(S_n^2/\sigma^2 - 1) \xrightarrow{d} N(0, 2)$. The delta method with $g(u) = \sqrt{u}$, $g'(1) = \frac{1}{2}$, gives $\sqrt{n-1}(S_n/\sigma - 1) \xrightarrow{d} N(0, \frac{1}{2})$; multiply by σ.

Solution 2024-05 #2

Since $X_i^2 = \sigma^2$, $(n - 1)S_n^2 = \sum X_i^2 - n\bar{X}^2 = n\sigma^2 - n\bar{X}^2$, so $S_n^2 - \sigma^2 = \frac{\sigma^2 - n\bar{X}^2}{n-1}$. By the CLT, $n\bar{X}^2 = (\sqrt{n}\bar{X})^2 = O_p(1)$, so $S_n^2 - \sigma^2 = O_p(n^{-1})$. Then $|S_n - \sigma| = |S_n^2 - \sigma^2|/(S_n + \sigma) \le |S_n^2 - \sigma^2|/\sigma = O_p(n^{-1})$, which is $o_p((n - 1)^{-\gamma})$ for every $\gamma < 1$. The rate is faster than $n^{-1/2}$ because $\mu_4 = \sigma^4$ makes the limit variance $(\mu_4 - \sigma^4)/(4\sigma^2)$ vanish.

Solution 2024-05 #3

$L(\theta; \mathbf{x}) = \mathbf{1}\{x_{(n)} - 1 < \theta < x_{(1)}\}$. For two samples, $L(\theta; \mathbf{x})/L(\theta; \mathbf{y})$ is free of θ iff the two intervals $(x_{(n)} - 1, x_{(1)})$ and $(y_{(n)} - 1, y_{(1)})$ coincide, i.e. iff $(x_{(1)}, x_{(n)}) = (y_{(1)}, y_{(n)})$. Hence $(X_{(1)}, X_{(n)})$ is minimal sufficient (two-dimensional for a one-dimensional parameter).

Solution 2024-05 #4

$I - H$ annihilates $X\beta$, so $\mathbf{y}'(I - H)\mathbf{y}/\sigma^2 \sim \chi^2(n - p - 1)$, while $\mathbf{y}'H\mathbf{y}/\sigma^2 \sim \chi^2(p + 1; \delta)$ with $\delta = \beta' X' X\beta/\sigma^2$ (Theorem 10.11 of the notes). They are independent since $H(I - H) = 0$. Hence $\frac{n-p-1}{p+1}F \sim F(p + 1, n - p - 1; \delta)$, a central F only when $\beta = \mathbf{0}$.

Solution 2024-10 #1

With $k = (\theta_0/\theta_1)^n$, $p_{\theta_1} = kp_{\theta_0}$ on $\{x_{(n)} \le \theta_0\}$, and $p_{\theta_0} = 0 < p_{\theta_1}$ on $\{\theta_0 < x_{(n)} \le \theta_1\}$. For any ψ with $E_{\theta_0}\psi \le \alpha$,

$$E_{\theta_1}\phi - E_{\theta_1}\psi = \int_{x_{(n)} \le \theta_0} (\phi - \psi)kp_{\theta_0} + \int_{x_{(n)} > \theta_0} (1 - \psi)p_{\theta_1} \ge k(\alpha - E_{\theta_0}\psi) \ge 0.$$

Solution 2024-10 #2

See Problem 5(a). Since the cdf of Z is continuous, every $\pm k$ is a continuity point, and

$$\limsup_n P(|X_n| > k) = P(|Z| > k) \to 0 \qquad (k \to \infty).$$

The finitely many early n are handled by enlarging k.

Solution 2024-10 #3

Let (a_n) be Cauchy in a compact A. By sequential compactness a subsequence $a_{n_j} \to a \in A$. Given ϵ, choose N with $d(a_n, a_m) < \epsilon/2$ for $n, m \ge N$, and j with $n_j \ge N$ and $d(a_{n_j}, a) < \epsilon/2$. Then $d(a_n, a) < \epsilon$ for $n \ge N$, so $a_n \to a \in A$.

Solution 2025-04 #1

(1) $P(|X_n+Y_n|>2k)\le P(|X_n|>k)+P(|Y_n|>k)$. (2) $P(|X_nY_n|>k^2)\le P(|X_n|>k)+P(|Y_n|>k)$. Take $\sup_n$ and let $k\to\infty$ in both. (3) See Problem 5(b). (4) Given ϵ, choose N with $P(|Z_n|>1)<\epsilon$ for $n\ge N$. For each of the finitely many $n<N$, $P(|Z_n|>k)\to 0$ as $k\to\infty$. So $\sup_n P(|Z_n|>k)<\epsilon$ for all large k.

Solution 2025-04 #2

(1) $m_1=\frac{\alpha}{\alpha+1}$ gives $\alpha=\frac{m_1}{1-m_1}$, hence the MME. With $g(m)=\frac{m}{1-m}$, $g'(m_1)=(\alpha+1)^2$, and $\operatorname{Var}X=\frac{\alpha}{(\alpha+1)^2(\alpha+2)}$, so $\sqrt{n}(\hat\alpha^{\mathrm{MME}}-\alpha)\xrightarrow{d}N\big(0,\frac{\alpha(\alpha+1)^2}{\alpha+2}\big)$. (2) $\hat\alpha^{\mathrm{MLE}}=-n/\sum\log X_i$; $I(\alpha)=\alpha^{-2}$, so the limit is $N(0,\alpha^2)$. (3) $ARE=\frac{\alpha(\alpha+2)}{(\alpha+1)^2}<1$.

Solution 2025-04 #3

(1), (3): see Problem 11(a), using $W_n=2W_n^+-n(n+1)/2$. (2) With $\sigma_n^2=\sum j^2$, the cgf of W_n/σ_n is $\sum_j\log\cosh(jt/\sigma_n)=\frac{t^2}{2}+O(\sum_j j^4/\sigma_n^4)$, and $\sum j^4/\sigma_n^4=O(n^5/n^6)\to 0$. So the cgf tends to $t^2/2$.

Solution 2025-10 #1

(a) $L=2^n\prod x_i\,\theta^{-2n}\mathbf{1}\{\theta\ge x_{(n)}\}$ is decreasing in θ, so $\hat\theta^{\mathrm{MLE}}=X_{(n)}$. $EX=\frac{2}{3}\theta$ gives $\hat\theta^{\mathrm{MME}}=\frac{3}{2}\bar X$.

(b) $X_{(n)}$ has cdf $(x/\theta)^{2n}$: $EX_{(n)}=\frac{2n}{2n+1}\theta$, $EX_{(n)}^2=\frac{n}{n+1}\theta^2$, so $\operatorname{Var}X_{(n)}=\frac{n\theta^2}{(n+1)(2n+1)^2}$. $\hat\theta^{\mathrm{MME}}$ is unbiased, and with $\operatorname{Var}X=\frac{\theta^2}{2}-\frac{4\theta^2}{9}=\frac{\theta^2}{18}$, $\operatorname{Var}\hat\theta^{\mathrm{MME}}=\frac{9}{4}\cdot\frac{\theta^2}{18n}=\frac{\theta^2}{8n}$.

(c) $MSE(\hat\theta^{\mathrm{MLE}})=\frac{n\theta^2}{(n+1)(2n+1)^2}+\frac{\theta^2}{(2n+1)^2}=\frac{\theta^2}{(n+1)(2n+1)}$. This is below $\frac{\theta^2}{8n}$ iff $2n^2-5n+1>0$, i.e. $n\ge 3$. For $n=1,2$ the MME has smaller MSE; for $n\ge 3$ the MLE does, and its MSE decays like n^{-2}.

Solution 2025-10 #2

(a) $m(\mathbf{t})=\exp\{\frac{1}{2}(t_1^2+t_2^2+t_3^2)+\rho t_1t_2+\rho t_2t_3+\rho^2t_1t_3\}$. (b) With $\mathbf{a}=(1,-2,1)'$, $\Sigma\mathbf{a}=((1-\rho)^2,\,-2(1-\rho),\,(1-\rho)^2)'$, so $\mathbf{a}'\Sigma\mathbf{a}=2(1-\rho)^2+4(1-\rho)=2(1-\rho)(3-\rho)$. (c) $\operatorname{Cov}(X_1-X_2,X_2-X_3)=\rho-\rho^2-1+\rho=-(1-\rho)^2\neq 0$. They are jointly normal with nonzero covariance, hence dependent.

Solution 2025-10 #3

(a) If f is continuous on $[a,b]$ and differentiable on (a,b), there is $c\in(a,b)$ with $f'(c)=\frac{f(b)-f(a)}{b-a}$. Proof: $g(x)=f(x)-f(a)-\frac{f(b)-f(a)}{b-a}(x-a)$ has $g(a)=g(b)=0$. By the extreme value theorem g attains a maximum and a minimum on $[a,b]$; if both are at endpoints, $g\equiv 0$; otherwise an interior extremum c has $g'(c)=0$ (Rolle). (b) $f(x)=x^2$ on $[0,1]$: the chord slope is $1=f'(\frac{1}{2})$. (c) $f(x)=|x|$ on $[-1,1]$: the chord slope is 0, but $f'=\pm1$ wherever it exists. Differentiability fails at 0.

Solution 2026-04 #1

(a) The likelihood is $\theta_1^{x_1}(1-\theta_1)^{x_2+x_3}\theta_2^{x_2}(1-\theta_2)^{x_3}$, which separates, so $\hat\theta_1=X_1/n$ and $\hat\theta_2=X_2/(X_2+X_3)$. (b) For the n trials, $I=\operatorname{diag}\Big(\frac{n}{\theta_1(1-\theta_1)},\,\frac{n(1-\theta_1)}{\theta_2(1-\theta_2)}\Big)$, using $E(X_2+X_3)=n(1-\theta_1)$; the off-diagonal entries vanish because the log-likelihood is additive. (c) $\sqrt{n}(\hat\theta-\theta)\xrightarrow{d}N_2\Big(\mathbf{0},\operatorname{diag}\big(\theta_1(1-\theta_1),\frac{\theta_2(1-\theta_2)}{1-\theta_1}\big)\Big)$: the two estimators are asymptotically independent.

Solution 2026-04 #2

The density is $\frac{3}{4\pi}$ on the ball. (a) Integrating x_3 over a chord of length $2\sqrt{1-x_1^2-x_2^2}$ gives $f(x_1,x_2)=\frac{3}{2\pi}\sqrt{1-x_1^2-x_2^2}$ on the unit disk. (b) The cross-section is a disk of area $\pi(1-x_1^2)$, so $f(x_1)=\frac{3}{4}(1-x_1^2)$ on $[-1,1]$. (c) $f(x_1\mid x_2=0)\propto\sqrt{1-x_1^2}$, i.e. $\frac{2}{\pi}\sqrt{1-x_1^2}$ on $[-1,1]$.

(d) By symmetry the means and covariances vanish, and $\operatorname{Var} X_1 = \int_{-1}^{1} \frac{3}{4}x^2(1-x^2)dx = \frac{1}{5}$; so $\operatorname{Cov}(\mathbf{X}) = \frac{1}{5}I_3$.

Solution 2026-04 #3

(a) $f_n(0) = f_n(1) = 0$, and for $0 < x < 1$, $n(1-x)^n \to 0$; so the limit is 0. (b) $f_n'(x) = n(1-x)^{n-1}\{1-(n+1)x\}$, so the maximum is at $x = \frac{1}{n+1}$, with $f_n\big(\frac{1}{n+1}\big) = \big(\frac{n}{n+1}\big)^{n+1} \to e^{-1} \neq 0$. The convergence is not uniform.